THERMODYNAMICS AND THERMAL SYSTEMS (BAMEE203)

VIT®
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(Deemed to be University under section 3 of UGC Act, 1956)

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Syllabus

Module: 1	**Laws of Thermodynamics**	**9 hours**
Basic concepts; work and heat transfer; Zeroth law of thermodynamics; first law of thermodynamics - internal energy, working fluid - ideal gas; energy analysis of closed and open systems; second law of thermodynamics - heat engines, heat pumps, and refrigerators; reversible and irreversible processes; entropy; introduction to exergy		
Module: 2	**Gas and Vapour Power Cycles**	**9 hours**
Air standard cycles - Carnot, Otto, Diesel, Brayton cycles; properties of pure substances; property diagrams for phase change processes; vapor power cycles - Rankine cycle, reheat Rankine cycle, and regenerative Rankine cycle.		
Module: 3	**IC Engines**	**8 hours**
Introduction to IC engines; classification; testing and performance; simple carburetor; ignition system; fuel injection system - MPFI, CRDI, GDI; supercharging and turbocharging; combustion; knocking and detonation.		
Module: 4	**Air Compressor and Steam Nozzles**	**8 hours**
Reciprocating compressors - construction, working, effect of clearance volume, volumetric efficiency, isothermal efficiency, multi-staging; steam nozzles - one-dimensional steady flow through convergent-divergent nozzle; metastable flow.		
Module: 5	**Refrigeration and Air Conditioning**	**9 hours**
Vapor compression refrigeration system - components, working, P-h and T-s diagrams, calculation of COP, effect of sub-cooling and superheating, selection and properties of refrigerants; vapor absorption system; psychrometry - properties, processes, and chart; heating and cooling load calculations; air-conditioning systems.		

Course Objectives and Outcomes

Course Objectives

1. To explain the basic concepts of thermodynamics, including work, heat transfer, and energy analysis of systems.
2. To teach gas and vapor power cycles, IC engines, compressors, and nozzles for practical energy applications.
3. To guide the understanding of refrigeration, air-conditioning, and modern energy systems for evaluating performance and efficiency.

Course Outcomes (COs)

1. Apply the laws of thermodynamics to solve energy-related problems in closed and open systems.
2. Analyze gas and vapor power cycles to determine efficiency and performance
3. Calculate the performance of IC engines, compressors, and nozzles using standard principles
4. Apply refrigeration and air-conditioning principles to calculate system performance and design parameters
5. Assess contemporary energy systems and related technologies for practical engineering applications.

Text and Reference Books

Text-Books

1. Yunus A. Cengel, Michael A. Boles and Mehmet Kanoglu, Thermodynamics: An Engineering Approach, 2019, 9th Edition, McGraw Hill Education.
2. Nag P. K., Engineering Thermodynamics, 2017, 6th Edition, McGraw Hill Education.
3. Rajput R.K, "Thermal Engineering", Laxmi Publications Pvt Ltd, 10th Edition, 2017

Reference Books

1. Eastop T. D, McConkey A, "Applied Thermodynamics for Engineering Technologists", Pearson Education Ltd, 5th Edition, 2009
2. Ganesan, V., "Internal combustion engines", McGraw Hill Education Pvt Ltd, 1st Edition, 2012
3. Manohar Prasad, "Refrigeration and Air Conditioning", New Age International, 3rd Edition, 2015

Evaluation Method & Assignment Details

Component	Coverage	Weightage	Relevant CO's	Max. Marks
CAT - 1	Module 1-2	15%	CO1, CO2	50
CAT - 2	Module 2, 4 (Open Notes)	15%	CO2, CO4	50
Assignments	DA / Quiz / Project Review	30%	CO1 – CO5	30
FAT	2 Questions from each module.	40%	CO1-CO5	100
Total		**100%**		**230**

Assignment	Max. Marks
Assignment 1 (DA1)	10
Assignment 2 (DA2)	10
Assignment 3 (DA3)	10
Assignments (DA)	**30**

Module 2

Gas and Vapour Power Cycles

CO-2

Gas and Vapour and Power Cycles 9 hours

- Air standard cycles - Carnot, Otto, Diesel, Brayton cycles
- Properties of pure substances
- Property diagrams for phase change processes
- Vapor power cycles - Rankine cycle, reheat Rankine cycle, and regenerative Rankine cycle

Air Standard Cycles

- In Air Standard Cycles, the working fluid remains air throughout the cycle
- The energy is provided by burning a fuel
- The composition of the working fluid changes from air and fuel to combustion products during the cycle
- As air consists mostly of Nitrogen which does not react, the working fluid closely resembles air

AIR STANDARD – CARNOT CYCLE

1 THE CARNOT CYCLE – OVERVIEW

The best-known reversible cycle is the **Carnot Cycle**, proposed by **Sadi Carnot**.

The theoretical heat engine that operates on the Carnot Cycle is called the **Carnot Heat Engine**.

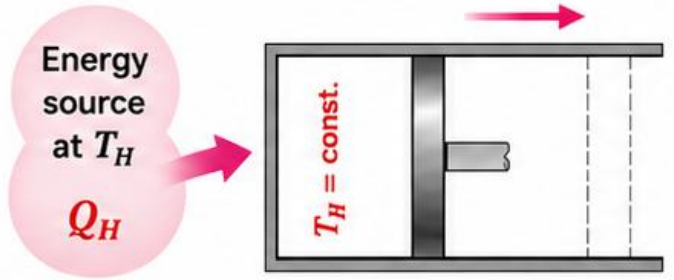
It consists of four reversible processes, **two isothermal** and **two adiabatic**.

KEY FEATURES

- The most efficient cycle between a source at T_H and a sink at T_L.
- Reversible isothermal heat transfer is difficult to achieve → **Not practical.**
- Used as a benchmark to compare the performance of actual cycles.

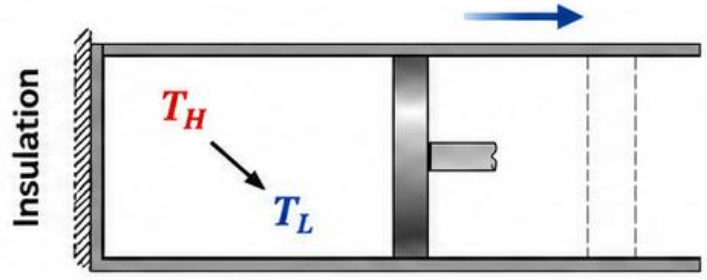
2 THE FOUR REVERSIBLE PROCESSES

1 REVERSIBLE ISOTHERMAL EXPANSION (1 → 2)



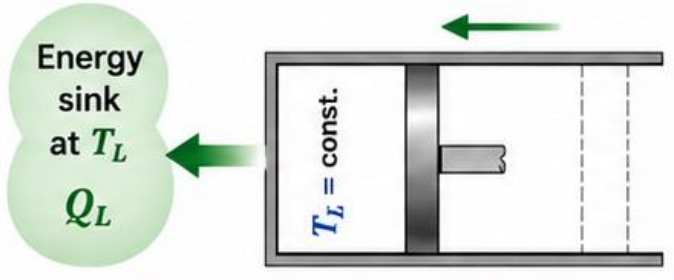
Heat absorbed at T_H

2 REVERSIBLE ADIABATIC EXPANSION (2 → 3)



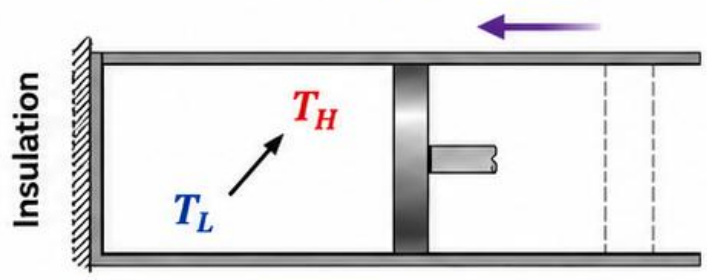
$Q = 0$ (Adiabatic)

3 REVERSIBLE ISOTHERMAL COMPRESSION (3 → 4)



Heat rejected at T_L

4 REVERSIBLE ADIABATIC COMPRESSION (4 → 1)



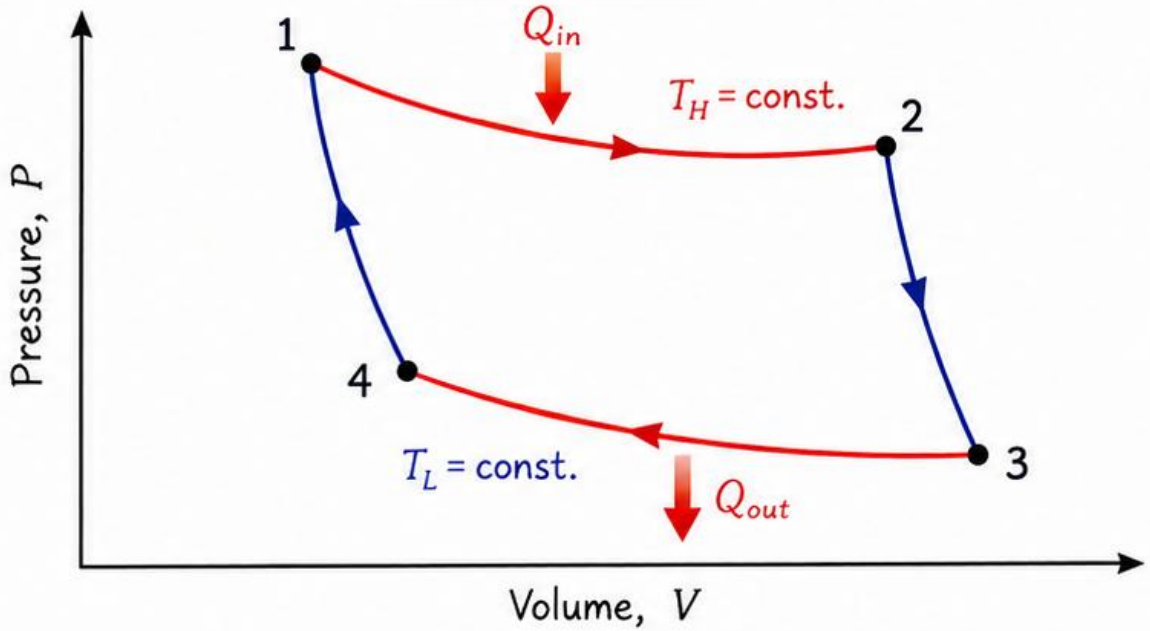
$Q = 0$ (Adiabatic)

Operates between two temperatures, T_H (high) and T_L (low), with heat interactions only at T_H and T_L and no other irreversibilities.

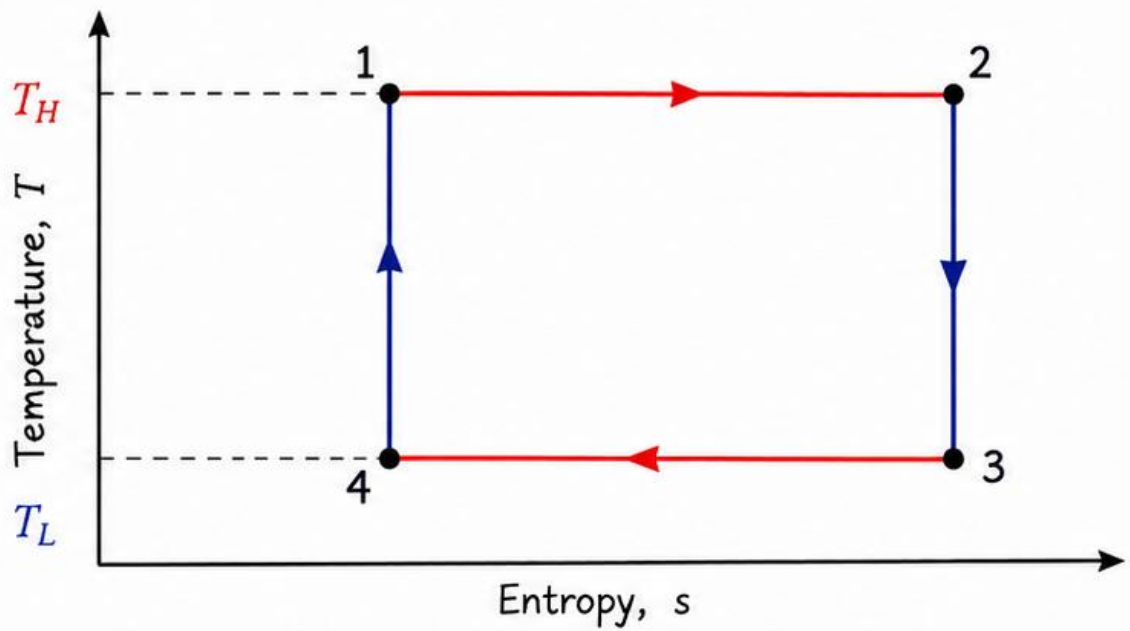
THE CARNOT CYCLE

The ideal reversible cycle consisting of two isothermal and two isentropic processes.

P-V DIAGRAM



T-S DIAGRAM



REVERSIBLE PROCESSES INVOLVED

1. Isothermal heat addition (1–2) at T_H — Heat absorbed: Q_{in}
2. Isentropic expansion (2–3)
3. Isothermal condensation (3–4) at T_L — Heat rejected: Q_{out}
4. Isentropic compression (4–1)

THERMAL EFFICIENCY OF CARNOT CYCLE

$$\eta = \frac{W_{net}}{Q_{supplied}} = 1 - \frac{T_3}{T_2} = 1 - \frac{T_L}{T_H}$$

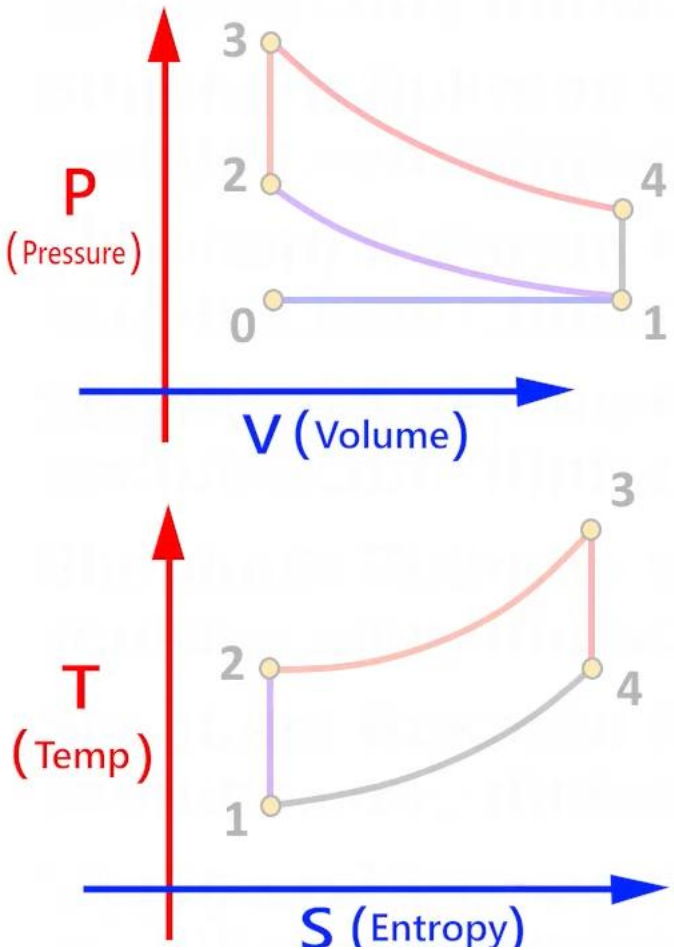
→ Q_{in} : Heat addition during isothermal expansion (1–2)

→ Q_{out} : Heat rejection during isothermal compression (3–4)

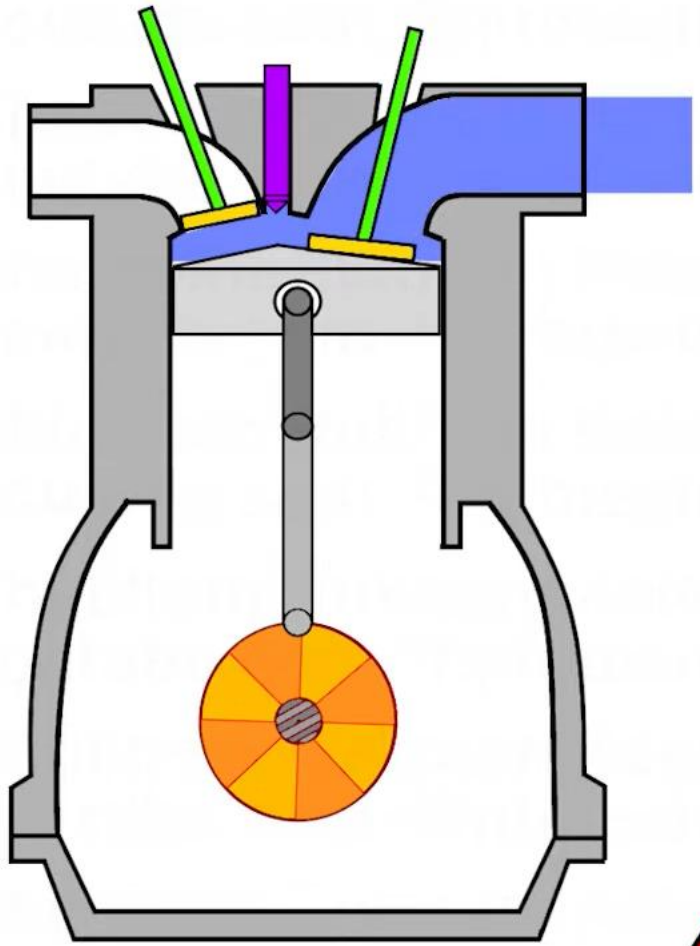
Otto Cycle

- an ideal thermodynamic cycle which makes your gasoline fuelled car to move and speed up by performing a four piston strokes of the car's engine.

Otto Cycle

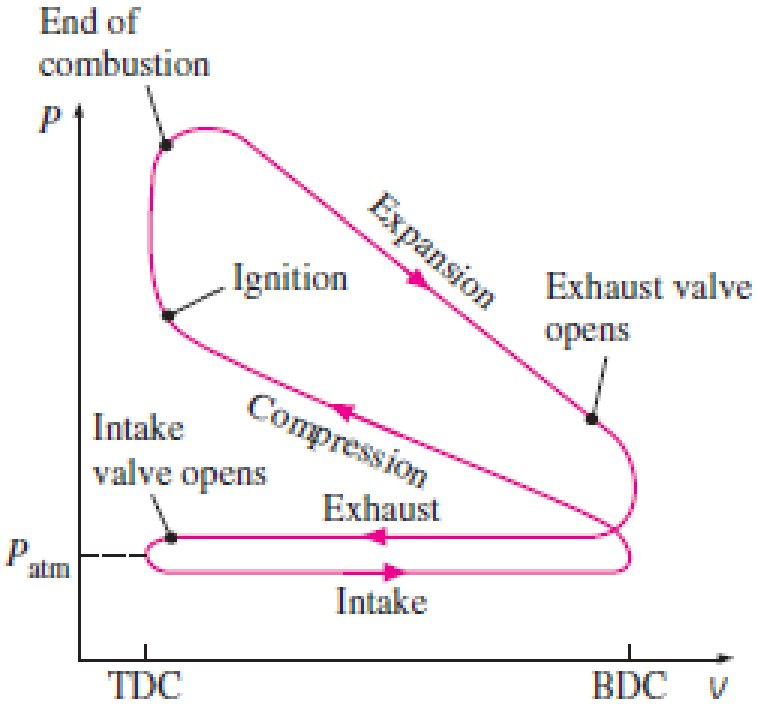


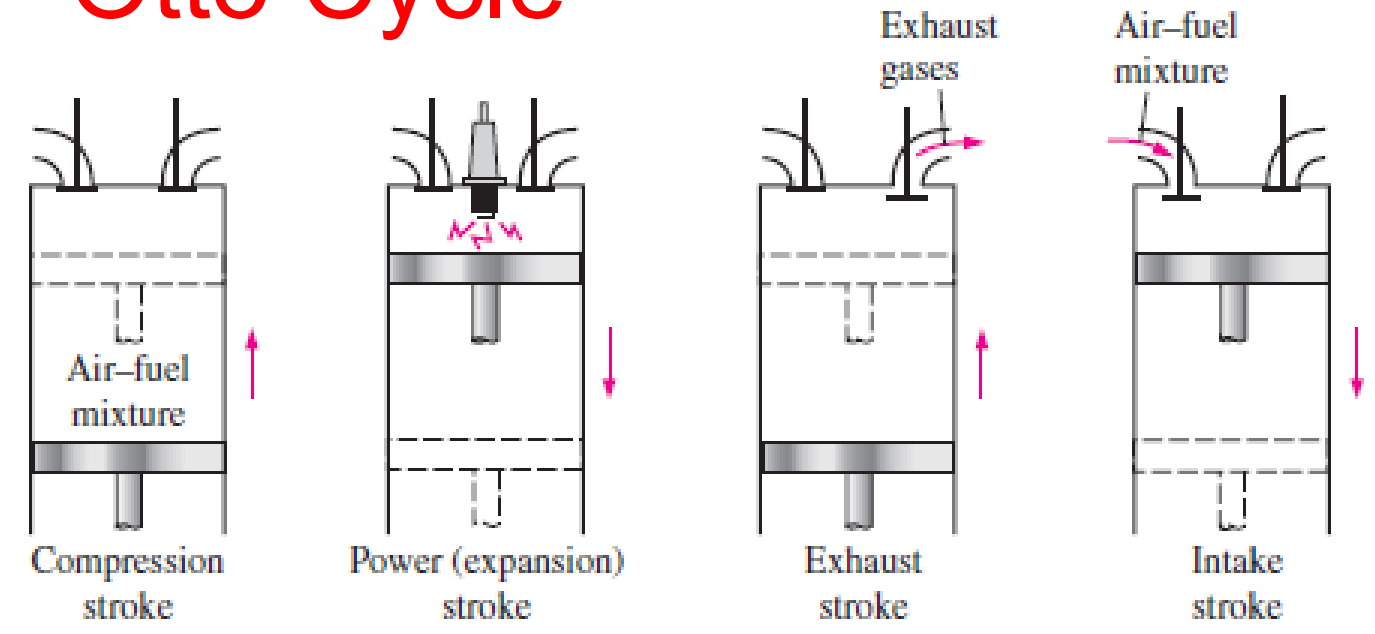
Process 0 to 1
Isobaric Process



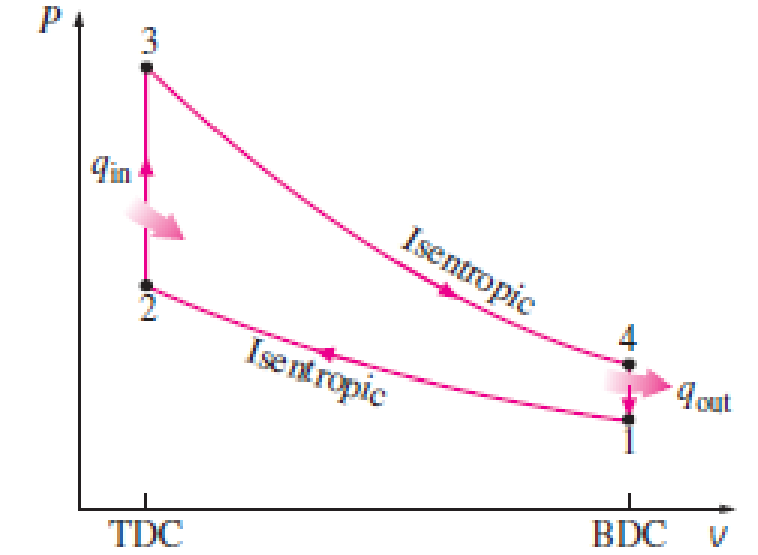
- Process 0 to 1
  Isobaric process
- Process 1 to 2
  Adiabatic Compression
- Process 2 to 3
  Isochoric process
- Process 3 to 4
  Adiabatic Expansion
- Process 4 to 1
  Isochoric process
- Process 1 to 0
  Isobaric process

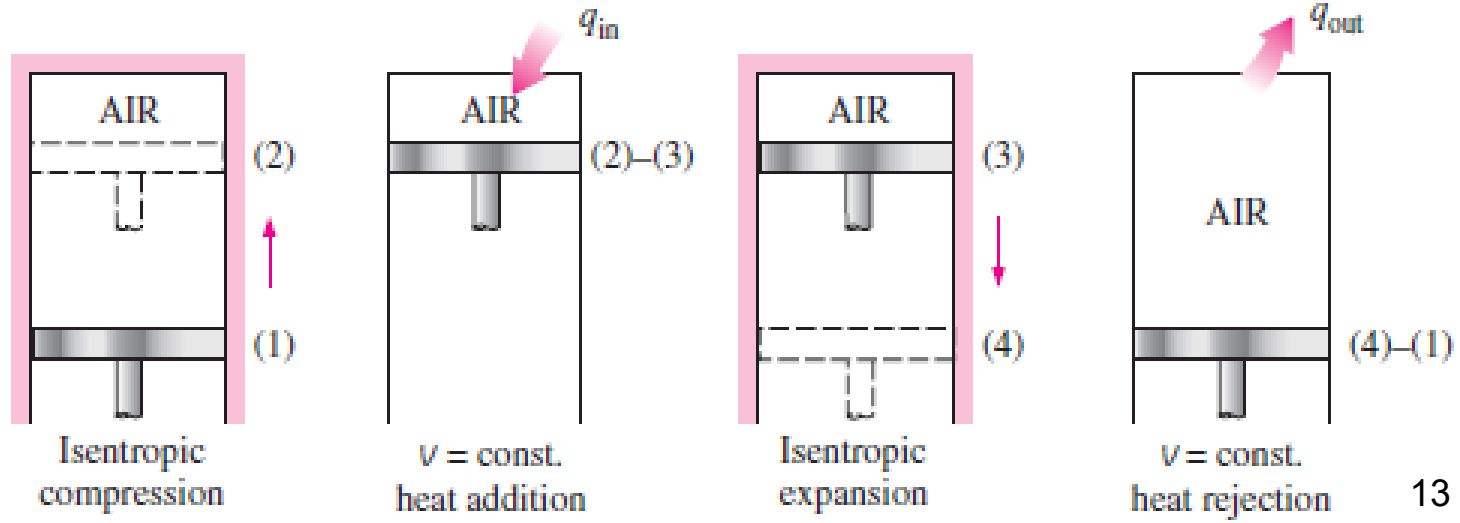
Otto Cycle





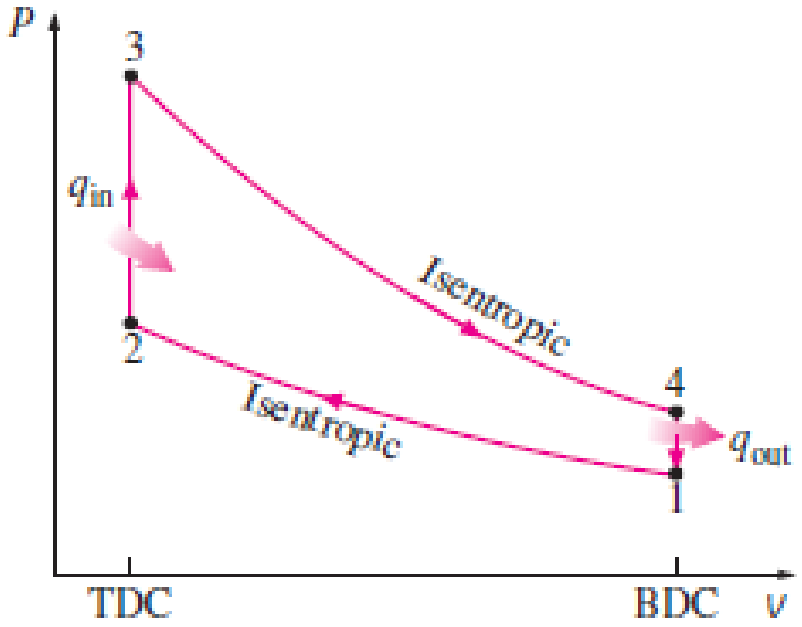
(*a*) Actual four-stroke spark-ignition engine

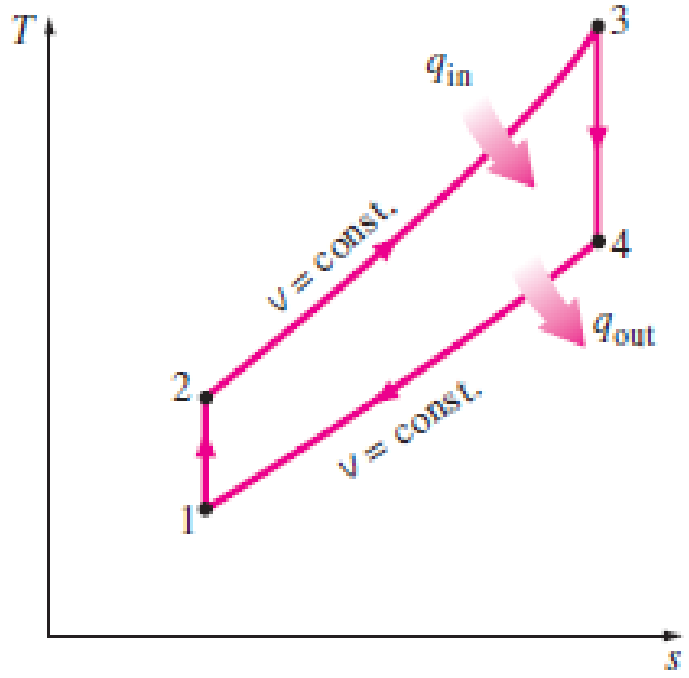




(*b*) Ideal Otto cycle

Otto Cycle





Processes:

1-2: Isentropic Compression

2-3: Constant Volume Heat Addition

3-4: Isentropic Expansion

4-1: Constant Volume Heat Rejection

Analysis of an Otto Cycle

- The Otto Cycle is executed in a closed system
- The changes in KE and PE are neglected
- No work is involved during the heat transfer processes

$$q-w=\Delta u$$

Thermodynamics Relations for an Ideal Gas undergoing Adiabatic Process

Adiabatic process:

$$PV^{\gamma} = C$$

$$P_1 V_1^{\gamma} = P_2 V_2^{\gamma} \quad \text{---(1)}$$

$$\boxed{\frac{V_1}{V_2} = \left(\frac{P_2}{P_1}\right)^{\gamma}}$$

$$PV = mRT$$

$$P = \frac{mRT}{V}$$

$$\Rightarrow \left(\frac{mRT}{V}\right) \cdot V^{\gamma} = C$$

$$TV^{\gamma-1} = \frac{C}{mR} \quad \Rightarrow \quad TV^{\gamma-1} = C$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\Rightarrow \boxed{\frac{V_1}{V_2} = \left(\frac{T_2}{T_1}\right)^{\gamma-1}} \quad \text{---(2)}$$

From equ. 1 & (2)

$$\left(\frac{P_2}{P_1}\right)^{\gamma} = \left(\frac{T_2}{T_1}\right)^{\gamma-1}$$

$$\Rightarrow \left(\frac{P_2}{P_1}\right)^{\frac{\gamma}{\gamma}} = \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$\boxed{\left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}}$$

Similarly,

$$\frac{T_1}{T_2} = \left(\frac{P_1}{P_2}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

Hence, for a reversible adiabatic process of an ideal gas, the following relations hold:

PV^{γ} = constant, $TV^{\gamma-1}$ = constant, and

$$\left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

Thermal Efficiency of Otto Cycle

1. Key Relations for Otto Cycle (Ideal Gas)

Heat addition (2 → 3) at constant volume:

$$q_{in} = u_3 - u_2 = C_v\,(T_3 - T_2)$$

Heat rejection (4 → 1) at constant volume:

$$q_{out} = u_4 - u_1 = C_v\,(T_4 - T_1)$$

2. Thermal Efficiency

$$\eta_{th,Otto} = \frac{W_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

$$= 1 - \frac{T_1\left(\frac{T_4}{T_1} - 1\right)}{T_2\left(\frac{T_3}{T_2} - 1\right)}$$

3. Using Adiabatic Relations

For isentropic (adiabatic) processes:

$$\frac{T_1}{T_2} = \left(\frac{v_2}{v_1}\right)^{\gamma-1} = \left(\frac{v_3}{v_4}\right)^{\gamma-1} = \frac{T_4}{T_3}$$

$$\eta_{th,Otto} = 1 - \frac{1}{r^{\gamma-1}}$$

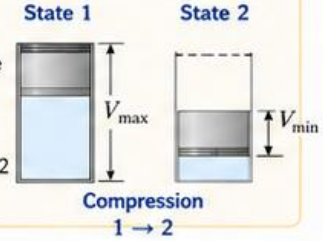
Compression Ratio

$$r = \frac{V_{max}}{V_{min}} = \frac{V_1}{V_2} = \frac{v_1}{v_2}$$

where V_{max}, V_1 = maximum (clearance) volume

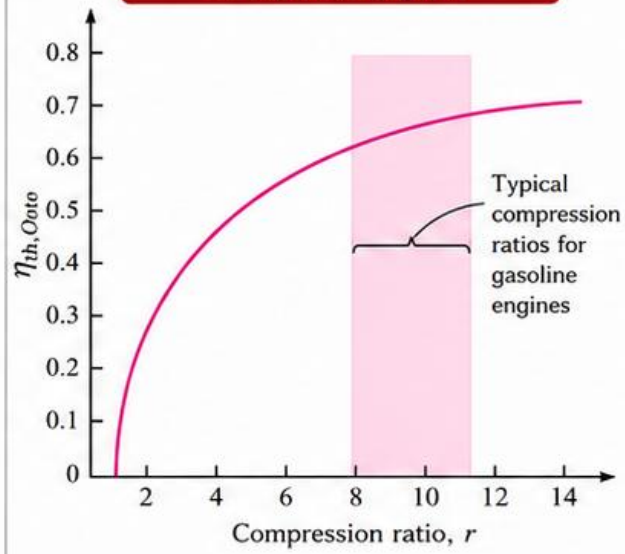
V_{min}, V_2 = minimum (cylinder) volume

v_1, v_2 = specific volumes at states 1 and 2

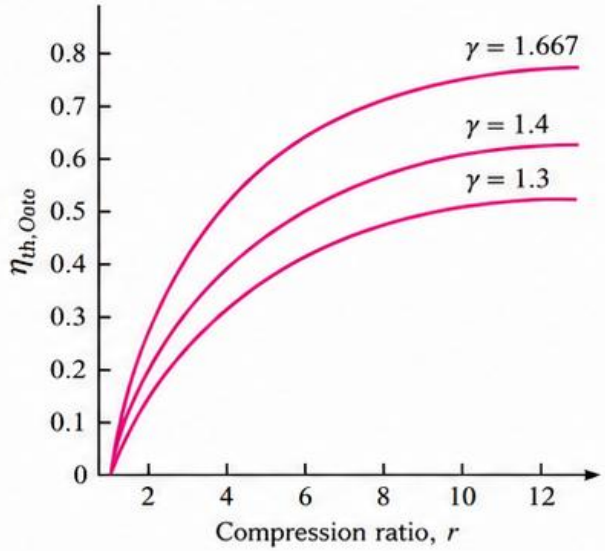


4. Variation of Efficiency with Compression Ratio (r)

(a) Typical Compression Ratios for Gasoline Engines



(b) Effect of Specific Heat Ratio (γ)



5. Key Takeaway

As the molecules of the working gas get larger, the **specific heat ratio (γ) decreases**. For a given compression ratio (r), this leads to **lower thermal efficiency**.

Hence, in actual engines (where molecules are larger and there are non-ideal effects), **the efficiency is lower** than the ideal Otto cycle.

Work Output

The net work output for an Otto cycle can be expressed as:

$$W = \frac{p_3V_3 - p_4V_4}{\gamma - 1} - \frac{p_2V_2 - p_1V_1}{\gamma - 1}$$

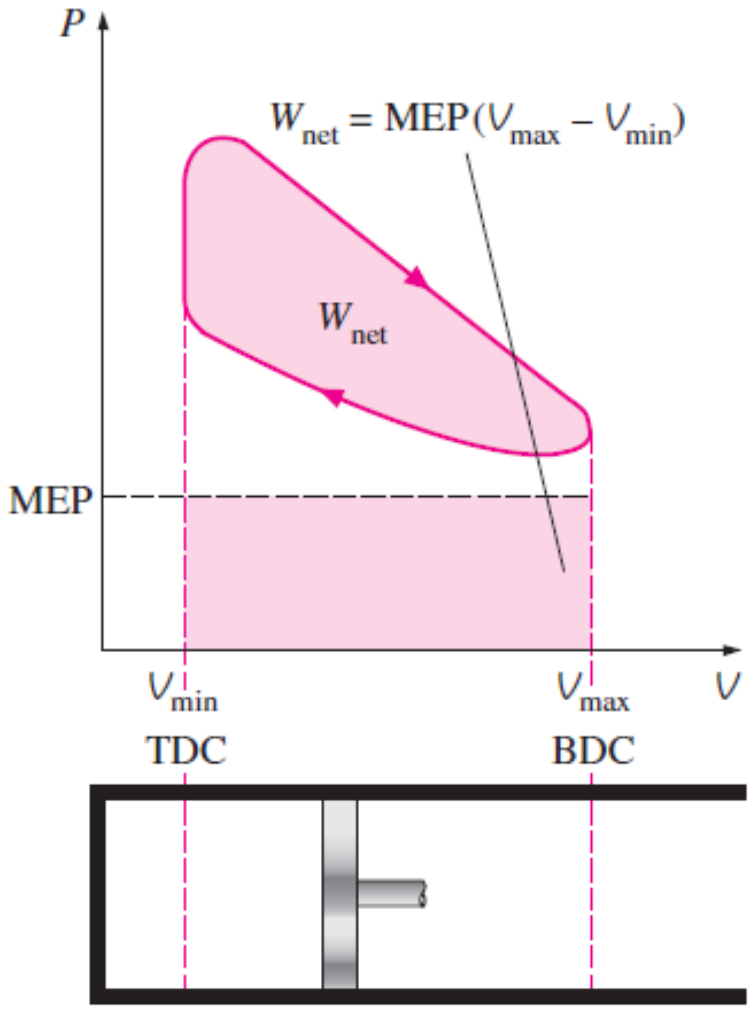
Also $\frac{p_2}{p_1} = \frac{p_3}{p_4} = r^\gamma$ and $\frac{p_3}{p_2} = \frac{p_4}{p_1} = r_p$ (say) Pressure Ratio

$$V_1 = rV_2 \quad \text{and} \quad V_4 = rV_3$$

therefore, $$W = \frac{p_1V_1}{\gamma - 1}\left(\frac{p_3V_3}{p_1V_1} - \frac{p_4V_4}{p_1V_1} - \frac{p_2V_2}{p_1V_1} + 1\right) = \frac{p_1V_1}{\gamma - 1}\left(\frac{r_pr^\gamma}{r} - r_p - \frac{r^\gamma}{r} + 1\right)$$

$$= \frac{p_1V_1}{\gamma - 1}\left(r_pr^{\gamma-1} - r_p - r^{\gamma-1} + 1\right) = \boxed{\frac{p_1V_1}{\gamma - 1}(r_p - 1)\left(r^{\gamma-1} - 1\right)}$$

Mean Effective Pressure (MEP)…



Mean Effective Pressure

The mean effective pressure of the cycle is given by:

$$p_m = \frac{\text{Work output}}{\text{Swept volume}}$$

$$\text{Swept volume} = V_1 - V_2 = V_2(r-1)$$

$$p_m = \frac{\frac{1}{\gamma-1}p_1V_1(r_p-1)\left(r^{(\gamma-1)}-1\right)}{V_2(r-1)} = \frac{p_1 r(r_p-1)\left(r^{(\gamma-1)}-1\right)}{(\gamma-1)(r-1)}$$

$$p_1V_1^{\gamma} = p_2V_2^{\gamma}$$

$$\frac{T_2}{T_1} = \frac{p_2V_2}{p_1V_1}$$

Mean Effective Pressure (MEP)

- Mean Effective Pressure is a fictitious pressure, if acting on the piston during the entire power stroke would produce the same amount of net work as that produced during the actual cycle
- The MEP is a parameter used to compare the performances of reciprocating engines of equal size
- Engine with larger MEP delivers more work per cycle and hence better

Problem - 9

In an Otto cycle, air at 17 ◦C and 1 bar is compressed adiabatically until the pressure is 15 bar. Heat is added at constant volume until the pressure rises to 40 bar. Calculate the compression ratio, air-standard efficiency and the mean effective pressure for the cycle.

Assume $c_V = 0.717$ kJ/kg K and $R = 8.314$ kJ/kmol K.

Compression Ratio

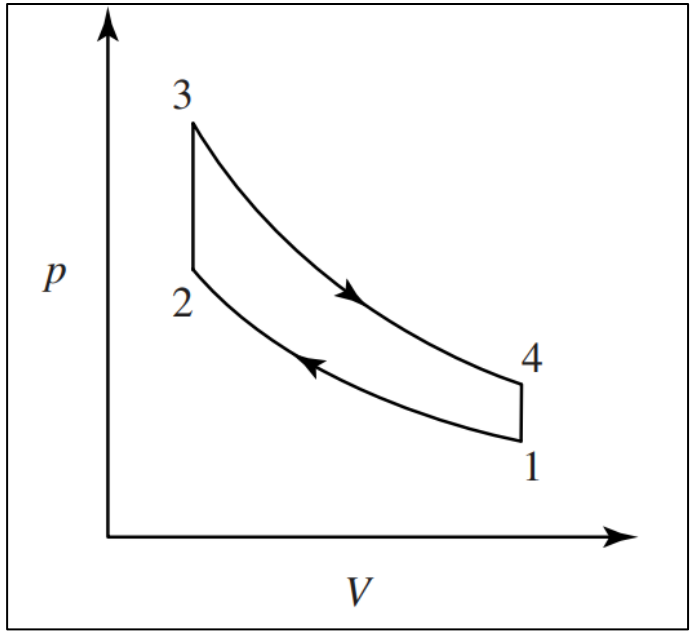
Process 1–2 is a reversible adiabatic (isentropic) compression, for which:

$$p_1 v_1^{\gamma} = p_2 v_2^{\gamma}$$

Rearranging for the compression ratio r = v_1 / v_2:

$$r = \frac{v_1}{v_2} = \left(\frac{p_2}{p_1}\right)^{\frac{1}{\gamma}} = \left(\frac{15}{1}\right)^{1/1.4} = 15^{0.7143}$$

r = 6.92



Air-Standard Efficiency

For the air-standard Otto cycle the thermal efficiency depends only on the compression ratio:

$$\eta_{otto} = 1 - \frac{1}{r^{\gamma-1}} = 1 - \frac{1}{(6.92)^{0.4}} = 1 - \frac{1}{2.168} = 0.5387$$

η = 53.87 %

Mean Effective Pressure

Isentropic compression 1–2:

$$T_2 = T_1 r^{\gamma-1} = 290 \times (6.92)^{0.4}$$

$$T_2 = 290 \times 2.168 = 628.7\ K$$

Constant-volume heat addition 2–3 ($v_2 = v_3$), so p/T is constant:

$$T_3 = T_2 \left(\frac{p_3}{p_2}\right)$$

$$T_3 = 628.7 \times \frac{40}{15} = 1676.5\ K$$

Heat supplied and net work

$$q_{in} = c_v(T_3 - T_2)$$

$$q_{in} = 0.717\ (1676.5 - 628.7)$$

$$q_{in} = 751.3\ kJ/kg$$

$$w_{net} = \eta \times q_{in}$$

$$w_{net} = 0.5387 \times 751.3 = 404.7\ kJ/kg$$

Swept (stroke) volume

$$v_1 = \frac{R\,T_1}{p_1} = \frac{0.287 \times 290}{100}$$

$$= 0.8323\ m^3/kg$$

$$v_1 - v_2 = v_1\left(1 - \frac{1}{r}\right)$$

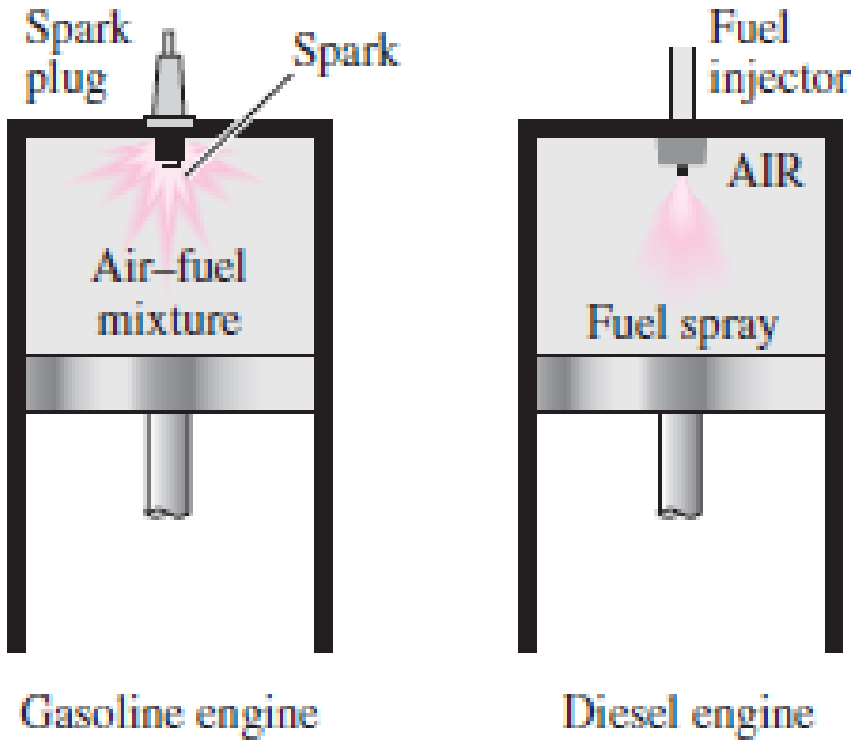
$$v_1 - v_2 = 0.8323\left(1 - \frac{1}{6.92}\right) = 0.7120\ m^3/kg$$

Mean effective pressure

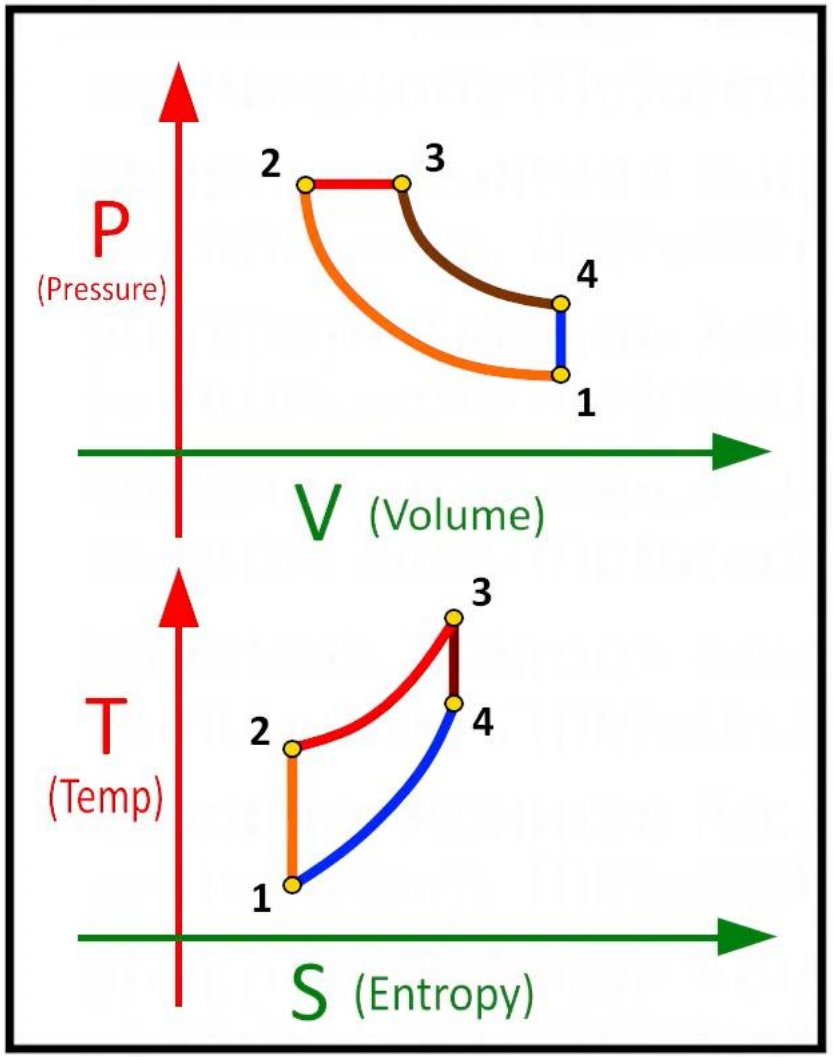
$$p_m = \frac{w_{net}}{v_1 - v_2}$$

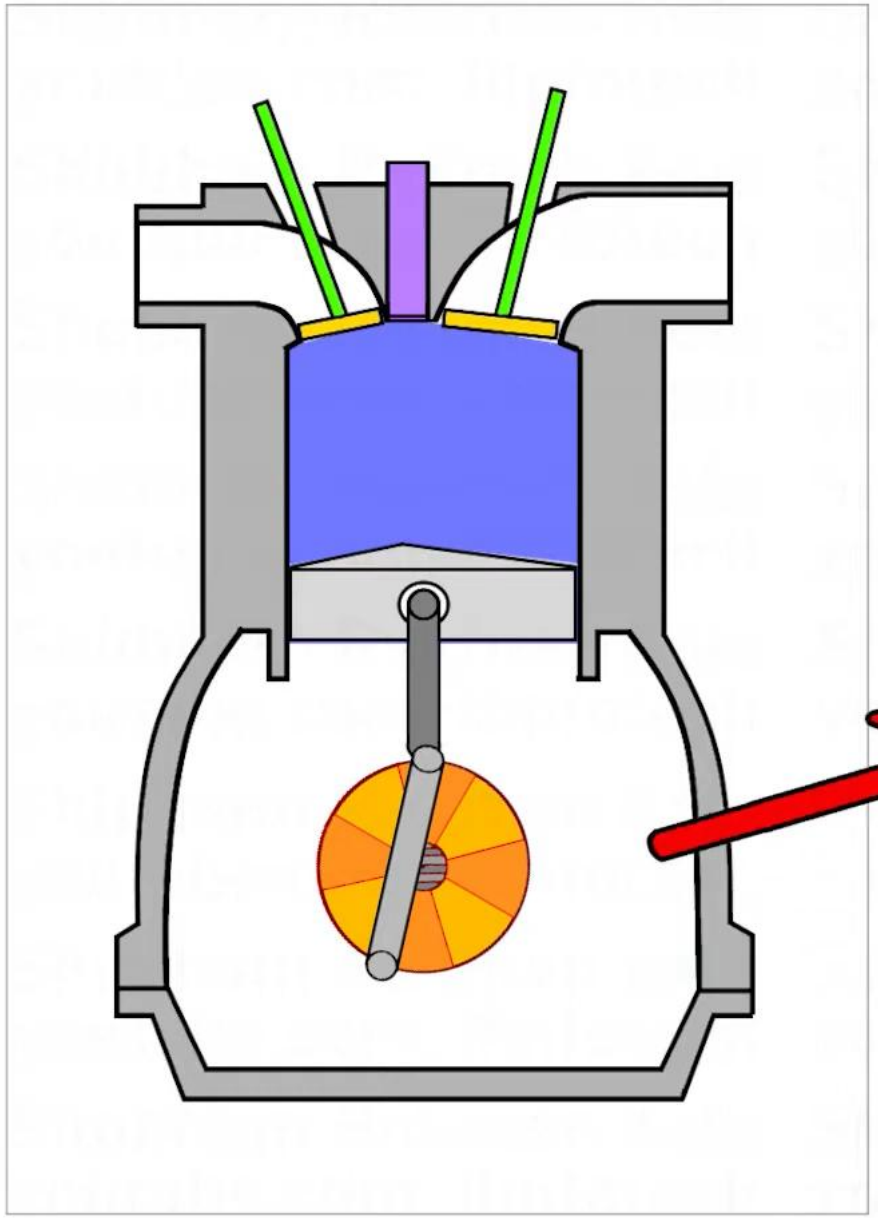
$$p_m = \frac{404.7}{0.7120} = 568.4\ kN/m^2 = 5.68\ bar$$

Diesel Engine and Differences from Petrol Engine



Diesel Cycle

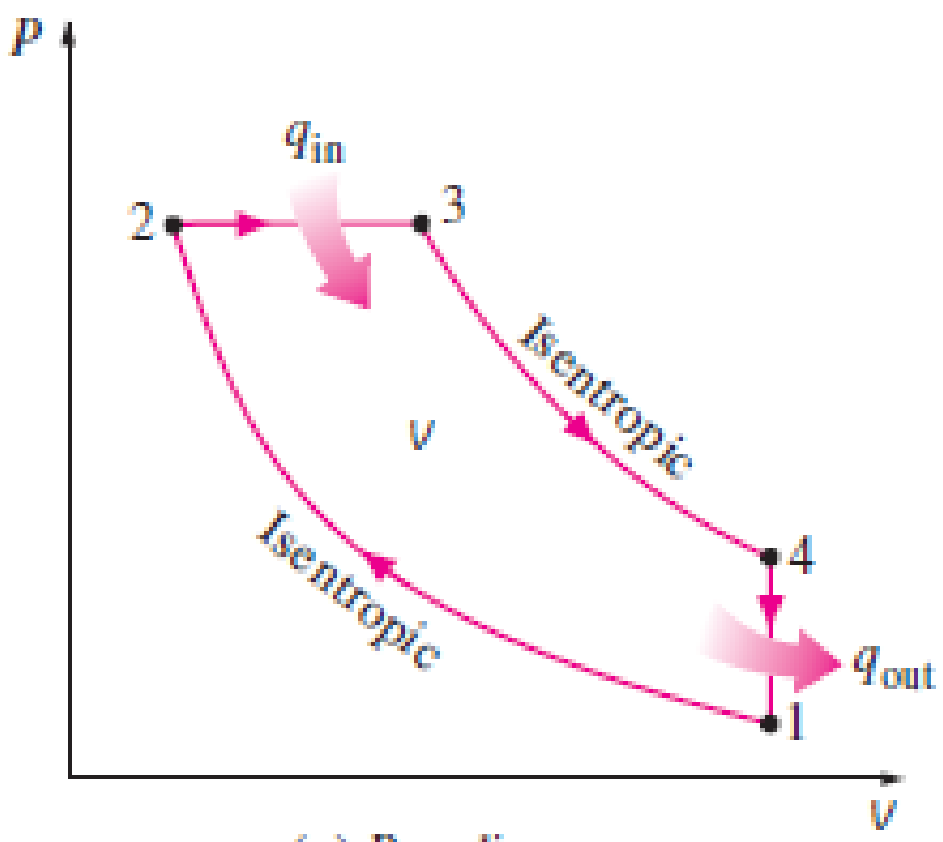




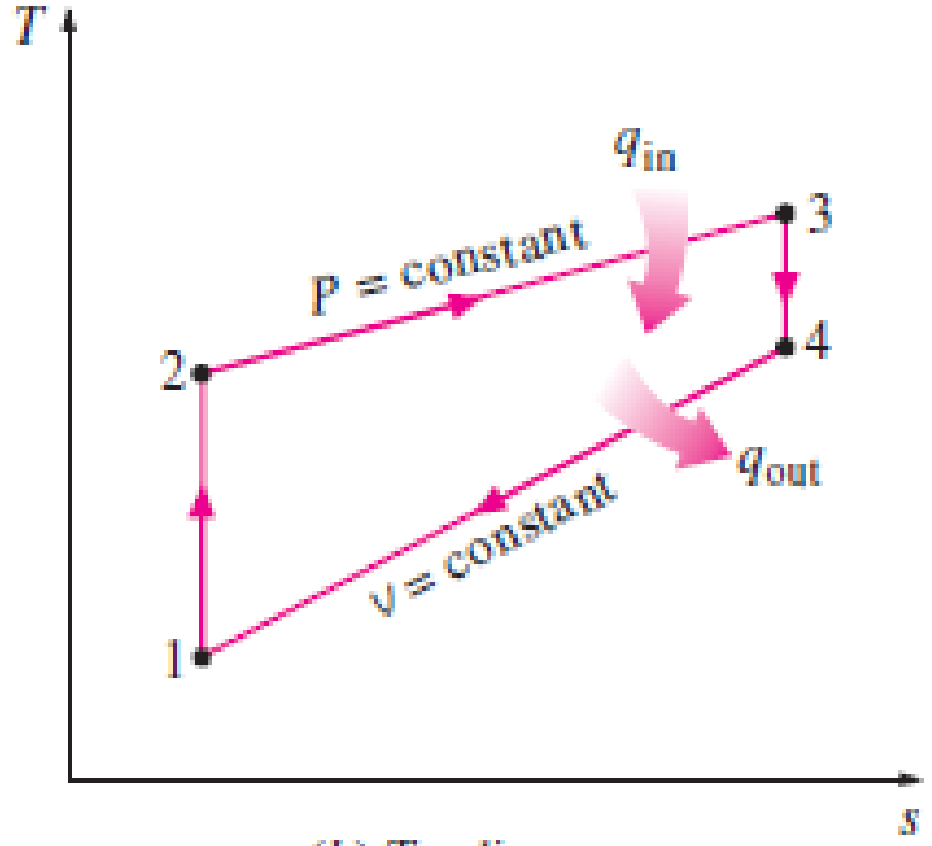
1. Process 1 to 2
   Isentropic Compression
2. Process 2 to 3
   Isobaric Heat Addition
3. Process 3 to 4
   Isentropic Expansion
4. Process 4 to 1
   Isochoric Expansion

YouTube/Ifinfotech

Diesel Cycle



(a) P- v diagram



(b) T-s diagram

Analysis of Diesel Cycle

$$q_{in} - w = u_3 - u_2$$

$$q_{in} = w + u_3 - u_2 = P_2\left(v_3 - v_2\right) + \left(u_3 - u_2\right) = h_3 - h_2 = C_p\left(T_3 - T_2\right)$$

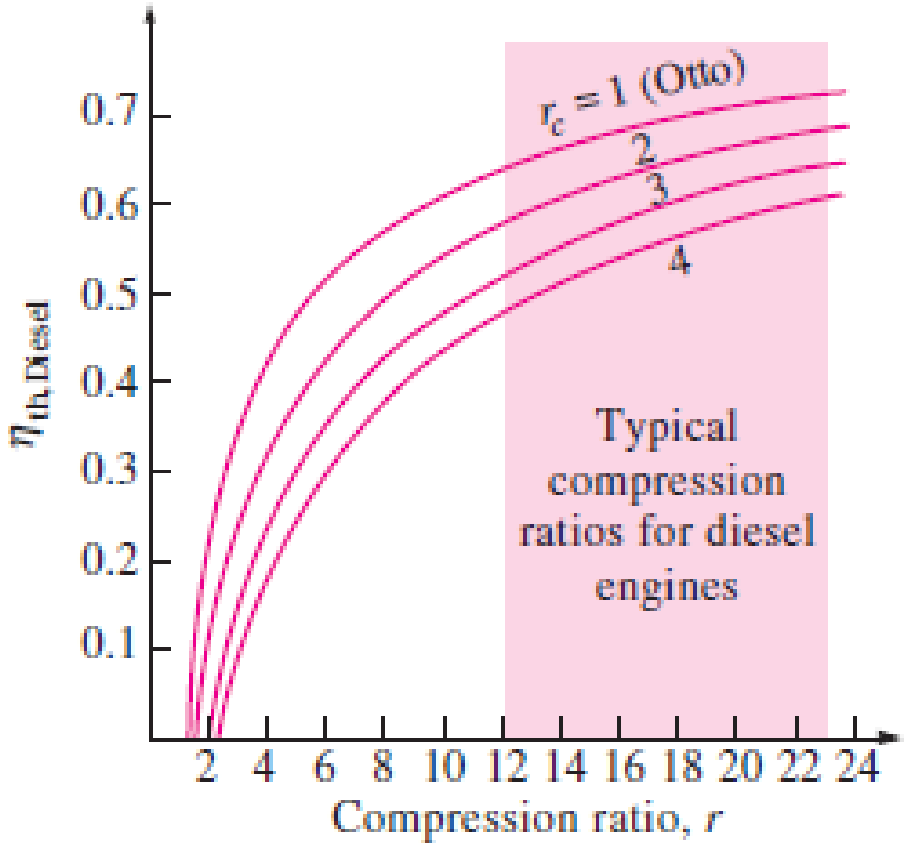
$$q_{out} = u_4 - u_1 = C_v\left(T_4 - T_1\right)$$

$$\eta_{th,Diesel} = \frac{W_{net}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{T_4 - T_1}{\gamma\left(T_3 - T_2\right)} = 1 - \frac{T_1\left(\frac{T_4}{T_1} - 1\right)}{\gamma T_2\left(\frac{T_3}{T_2} - 1\right)}$$

$$Cutoff\ \ Ratio = r_c = \frac{v_3}{v_2}$$

$$\eta_{th,Diesel} = 1 - \frac{1}{r^{\gamma-1}}\left[\frac{r_c^{\gamma} - 1}{\gamma\left(r_c - 1\right)}\right]$$

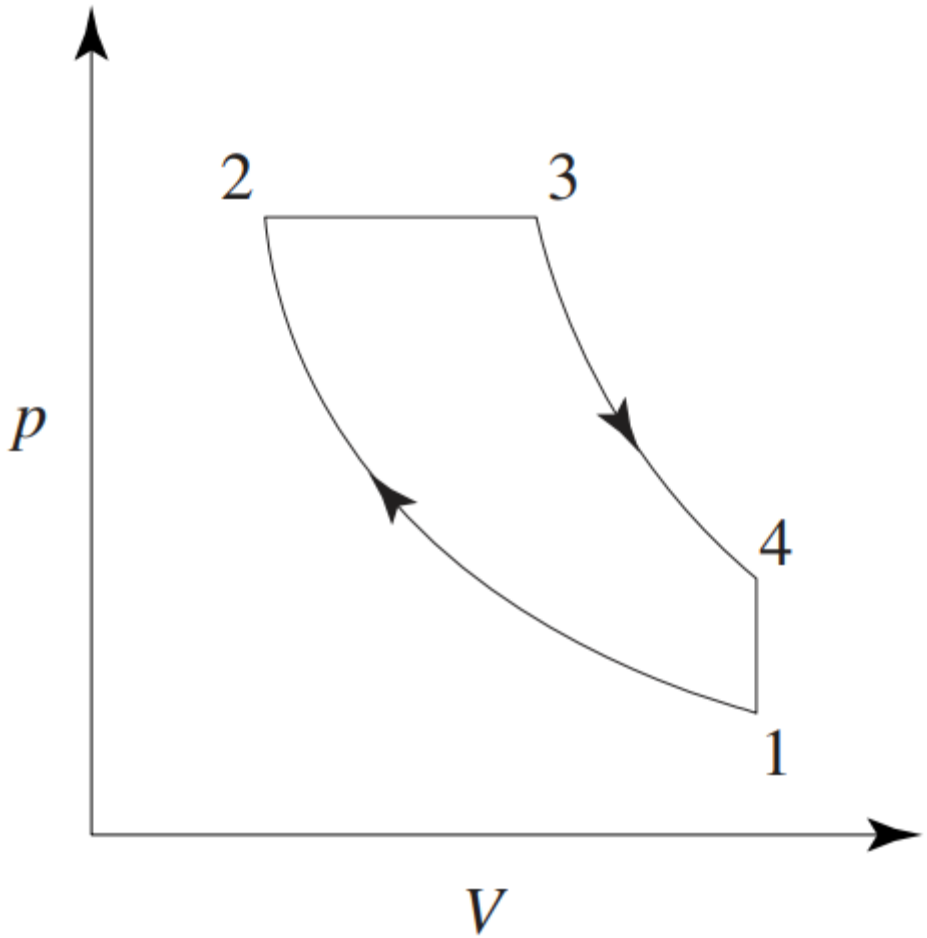
Efficiency of Diesel Cycle



For the same compression ratio, Otto Cycle is more efficient than the Diesel Cycle

Problem - 11

The mean effective pressure of an ideal Diesel cycle is 8 bar. If the initial pressure is 1.03 bar and the compression ratio is 12, determine the cut-off ratio and the air-standard efficiency. Assume ratio of specific heats for air to be 1.4.



$$\text{Work output} = p_m \times V_s = \text{ Area } 1234$$

$$= \text{Area under } 2-3 + \text{Area under } 3-4 - \text{Area under } 2-1$$

$$= p_2 (V_3 - V_2) + \frac{p_3 V_3 - p_4 V_4}{\gamma - 1} - \frac{p_2 V_2 - p_1 V_1}{\gamma - 1}$$

$$r = \frac{V_1}{V_2} = 1 + \frac{V_s}{V_c} = 12$$

$$V_s = 11 V_c \quad \text{and} \quad V_2 = V_c$$

$$V_1 = V_4 = 12 V_2 = 12 V_c$$

$$V_3 = r_c V_2 = r_c V_c$$

$$\frac{p_2}{p_1} = r^\gamma = 12^{1.4} = 32.42$$

$$p_2 = 32.42 \times 1.03 \times 10^5 = 33.39 \times 10^5 \text{ N/m}^2 = p_3$$

$$\frac{p_3}{p_4} = \left(\frac{r}{r_c}\right)^{1.4} = \frac{12^{1.4}}{r_c^{1.4}} = \frac{32.42}{r_c^{1.4}}$$

$$p_4 = \frac{33.39}{32.42} \times r_c^{1.4} \times 10^5 = 1.03 r_c^{1.4} \times 10^5$$

$$\text{Area } 1234 = 33.39\,(r_c V_c - V_c) + \frac{33.39 \times r_c V_c - 1.03 r_c^{1.4} \times 12 V_c}{0.4} - \frac{33.39 \times V_c - 1.03 \times 12 V_c}{0.4} \times 10^5$$

$$p_m \times V_s = 8 \times 11 V_c \times 10^5$$

$$\text{Area } 1234 = p_m \times V_s$$

Substituting and simplifying, $0.672 r_c - 0.178 r_c^{1.4} = 1$

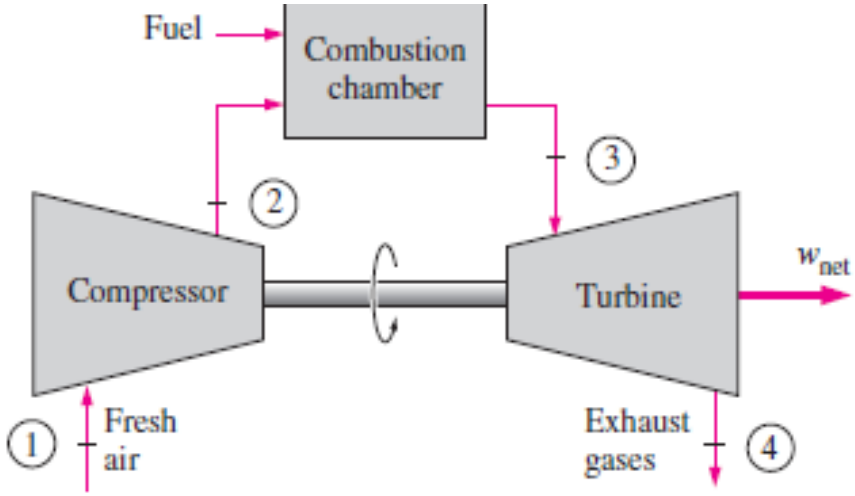
Solving by iteration,

$$r_c = 2.38$$

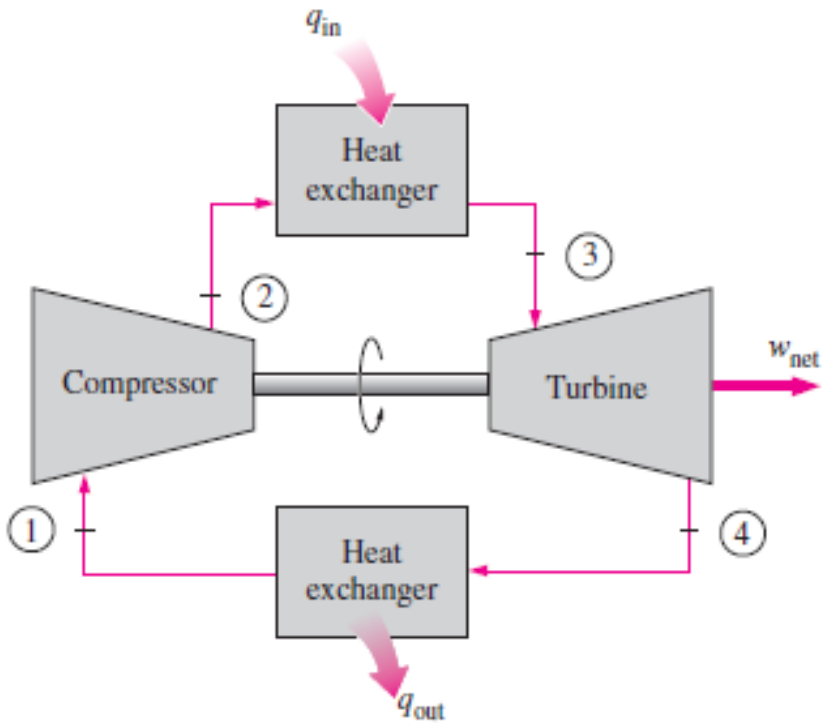
$$\eta = 1 - \frac{1}{r^{\gamma-1}} \frac{r_c^{\gamma} - 1}{\gamma(r_c - 1)} = 1 - \frac{1}{1.4 \times 12^{0.4}} \times \left(\frac{2.38^{1.4} - 1}{1.38} \right)$$

$$= 0.5466 = 54.66\%$$

Gas Turbine



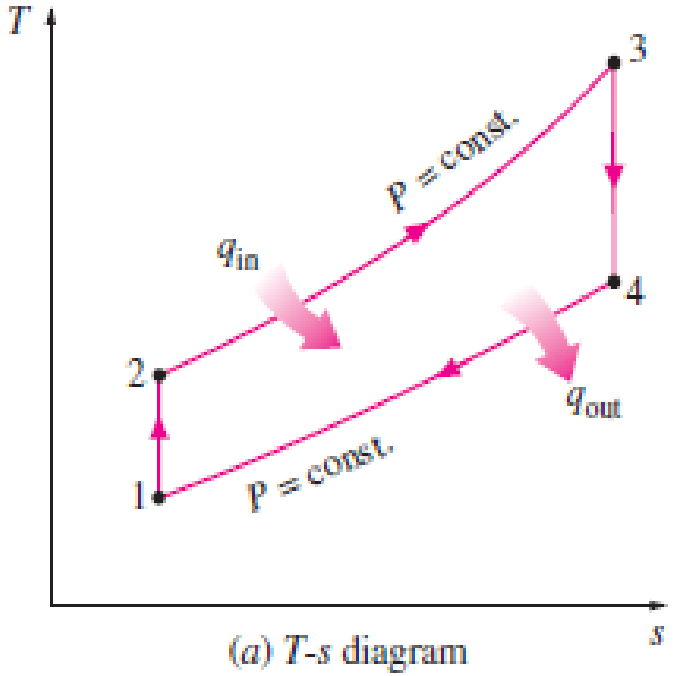
Open Cycle Gas Turbine



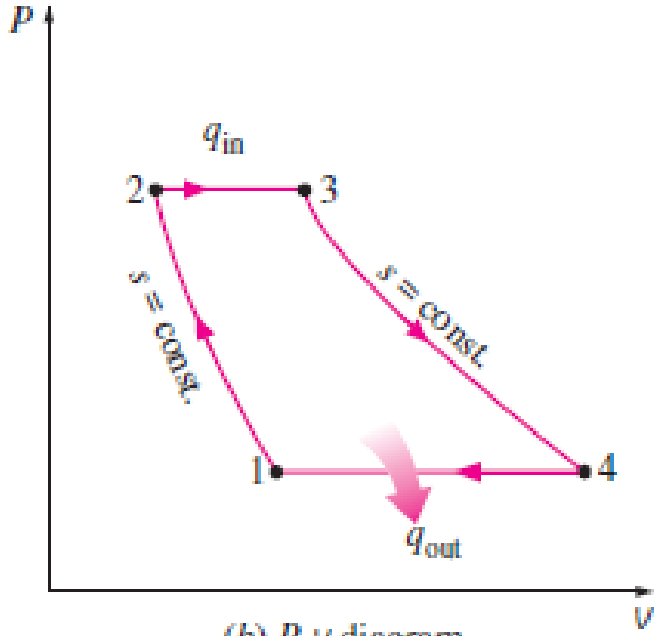
Closed Cycle

Gas Turbine

Brayton Cycle (Same as Rankine)



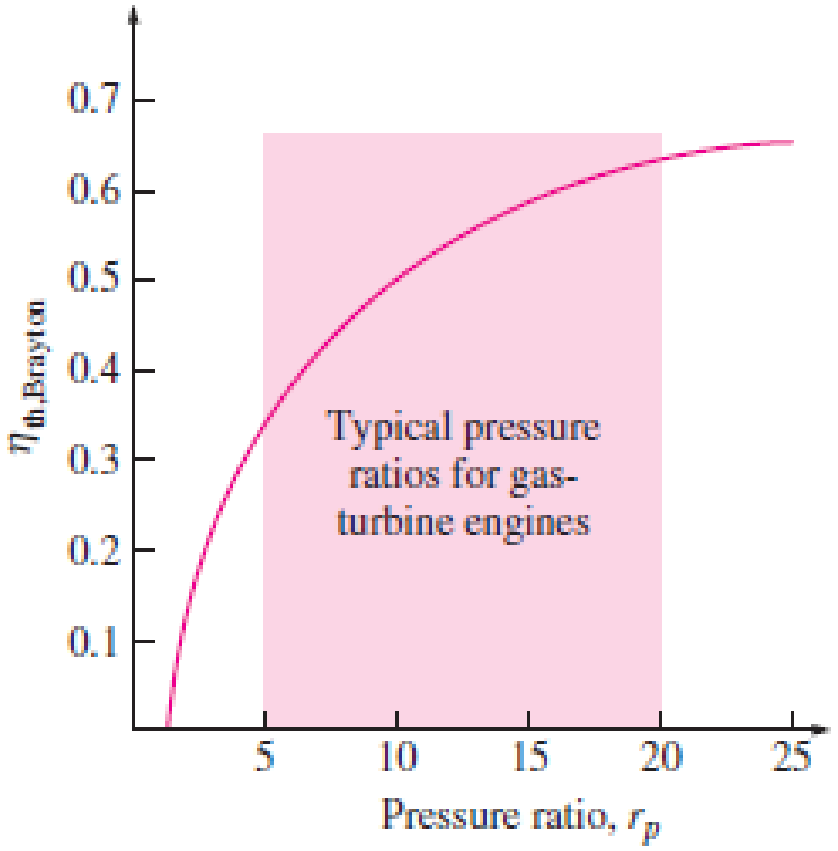
(a) T-s diagram



(b) P-v diagram

1-2: Isentropic Compression
2-3: Constant Pressure Heat addition
3-4: Isentropic Expansion
4-1: Constant Pressure Heat Rejection

Efficiency of Brayton Cycle



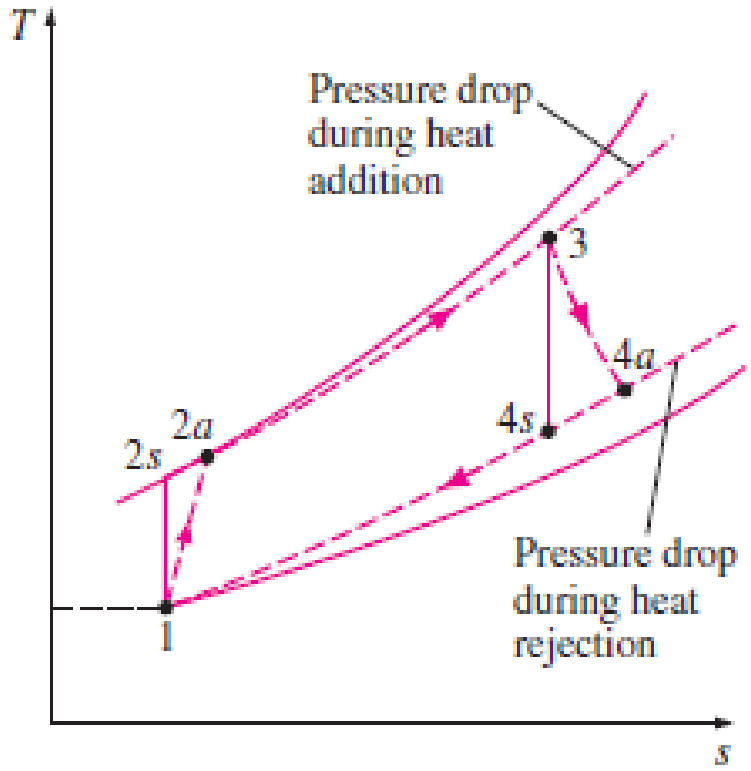
$$\eta_{th,Brayton} = 1 - \frac{1}{r_p^{(\gamma-1)/\gamma}}$$

$$\text{Pressure Ratio, } r_p = \frac{P_2}{P_1}$$

Back Work Ratio

- Back Work ratio in a gas turbine power plant is the ratio of the compressor work to the turbine work
- This ratio is very high
- More than half of the turbine work output is used to drive the compressor
- In steam power plants the back work ratio is very low

Isentropic Turbine and Compressor Efficiency



$$\eta_C = \frac{w_s}{w_a} = \frac{h_{2s} - h_1}{h_{2a} - h_1}$$

$$\eta_T = \frac{w_a}{w_s} = \frac{h_3 - h_{4a}}{h_3 - h_{4s}}$$

AIR STANDARD ASSUMPTIONS

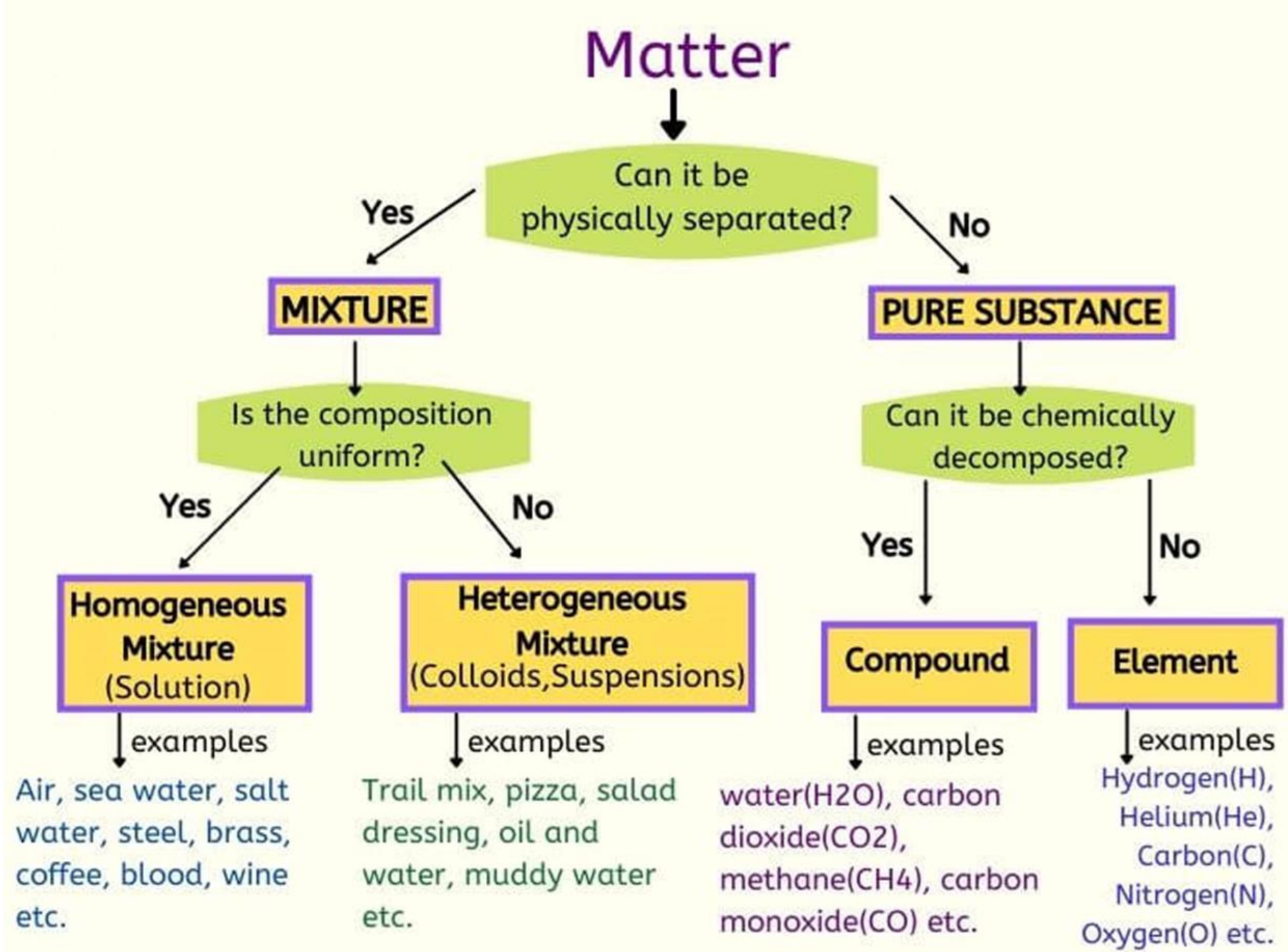
1. Air is considered as working fluid.
2. Air has constant specific heat.
3. Air behave as ideal gas.
4. All processes in cycle as internally reversible.
5. For air Change in K.E.= 0
6. For air Change in P.E.= 0
7. No internal heat generation.
8. heat is added & rejected externally.
9. The cycle is closed & processes are non-flow processes.
10. There is no frictional losses.

The physical constants, viz c_p, c_V, γ and M of working medium are the same as those of air at standard atmospheric conditions. For example, in SI units,

$c_p = 1.005$ kJ/kg K $\qquad M = 29$ kg/kmol

$c_V = 0.717$ kJ/kg K $\qquad \gamma = 1.4$

Properties of Pure Substances


Matter
Can it be physically separated?
Yes
No
MIXTURE
PURE SUBSTANCE
Is the composition uniform?
Can it be chemically decomposed?
Yes
No
Yes
No
Homogeneous Mixture (Solution)
Heterogeneous Mixture (Colloids,Suspensions)
Compound
Element
examples
examples
examples
examples
Air, sea water, salt water, steel, brass, coffee, blood, wine etc.
Trail mix, pizza, salad dressing, oil and water, muddy water etc.
water(H2O), carbon dioxide(CO2), methane(CH4), carbon monoxide(CO) etc.
Hydrogen(H), Helium(He), Carbon(C), Nitrogen(N), Oxygen(O) etc.

What is a Pure Substance ?

- A substance that has a fixed chemical composition throughout.
- A pure substance can exist in more than one phase, but its chemical composition must be the same in each phase.

Examples:

- Drinking water with ice cubes can be regarded as a pure substance because each phase has the same composition.
- A fuel-air mixture in the cylinder of an automobile engine can be regarded as a pure substance until ignition occurs.

Is pure substance comprised of two or more chemical element or compound?

- Can be single element: Such as, N_2, H_2, O_2
- Compound: Such as Water, H_2O, C_4H_{10},
- Mixture such as Air,
- 2-phase system such as H_2O.

A pure substance is comprised of a single chemical element or compound.

- Air is a mixture of several gases, but it is considered to be a pure substance.

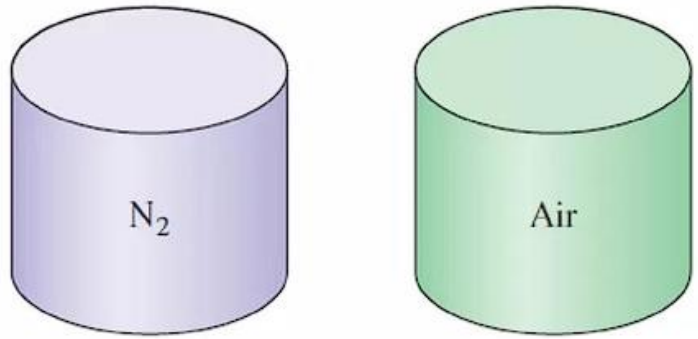
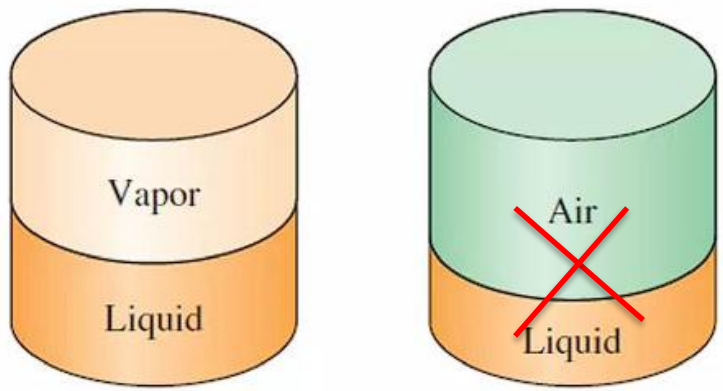


FIGURE 3–1
Nitrogen and gaseous air are pure substances.

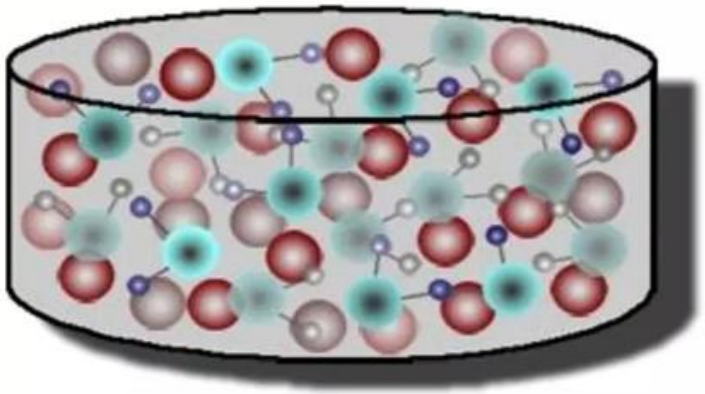


(*a*) H_2O (*b*) Air

FIGURE 3–2
A mixture of liquid and gaseous water is a pure substance, but a mixture of liquid and gaseous air is not.

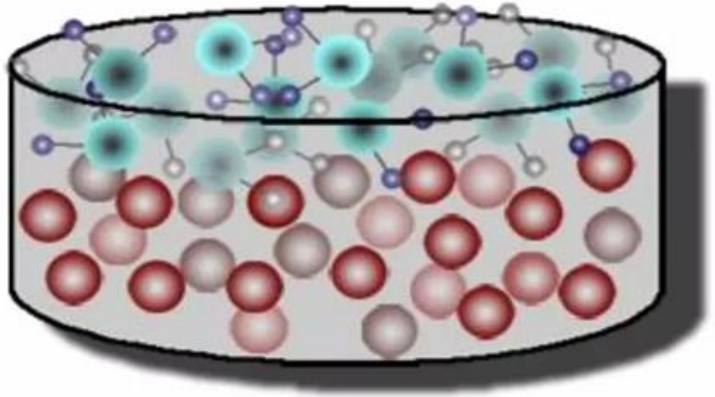
PURE SUBSTANCE

Homogeneous



- Because the mixture is **homogeneous** or uniform, the composition is the same throughout.
- As long as the composition remains constant and uniform throughout, this mixture can be treated as a pure substance. We call this a ***pseudocomponent.***

Heterogeneous



- Consider the **Heterogeneous** Mixture shown here. The composition near the top of the vessel is very different from the composition near the bottom of the vessel.

A **Heterogeneous** (or non-uniform)

Phases of a Pure Substance

PHASE:-

A quantity of matter that is homogeneous throughout in both chemical composition and physical structure.

Homogeneity in physical structure means that the matter is all *solid*, or all *liquid*, or all *vapor (gas)*.

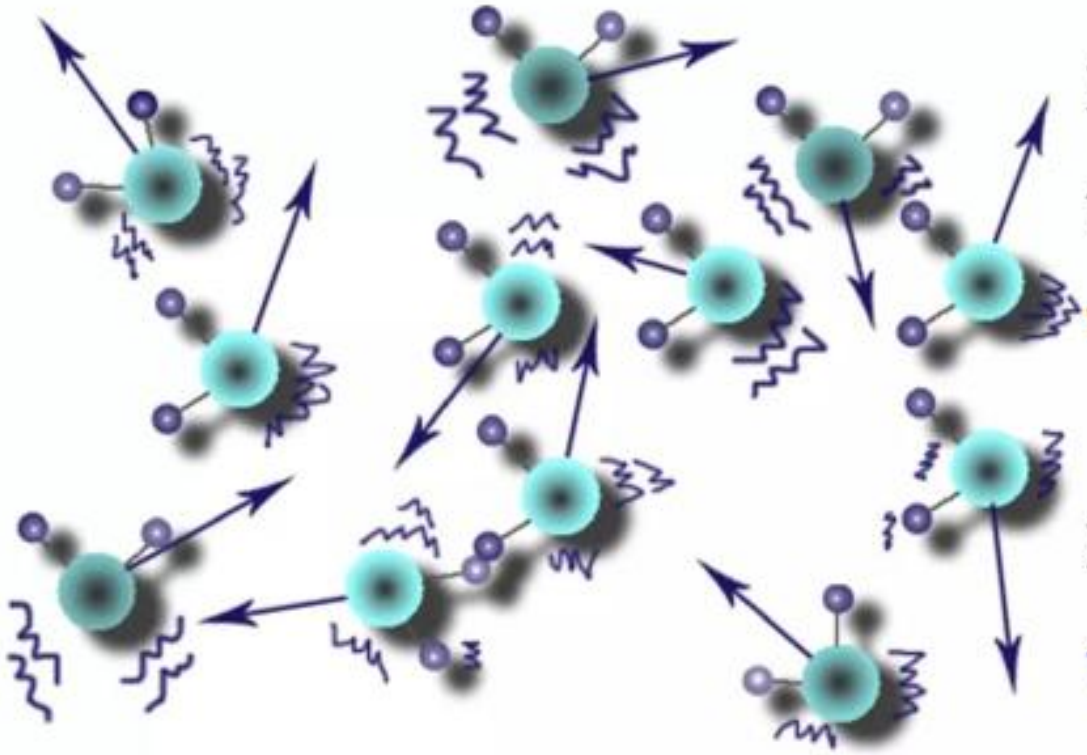
Examples:

► The air we breathe is a gas phase consisting of a mixture of different gases.

► Drinking water with ice cubes contains two phases of water: liquid and solid.

► Vinegar and olive oil salad dressing contains two different liquid phases.

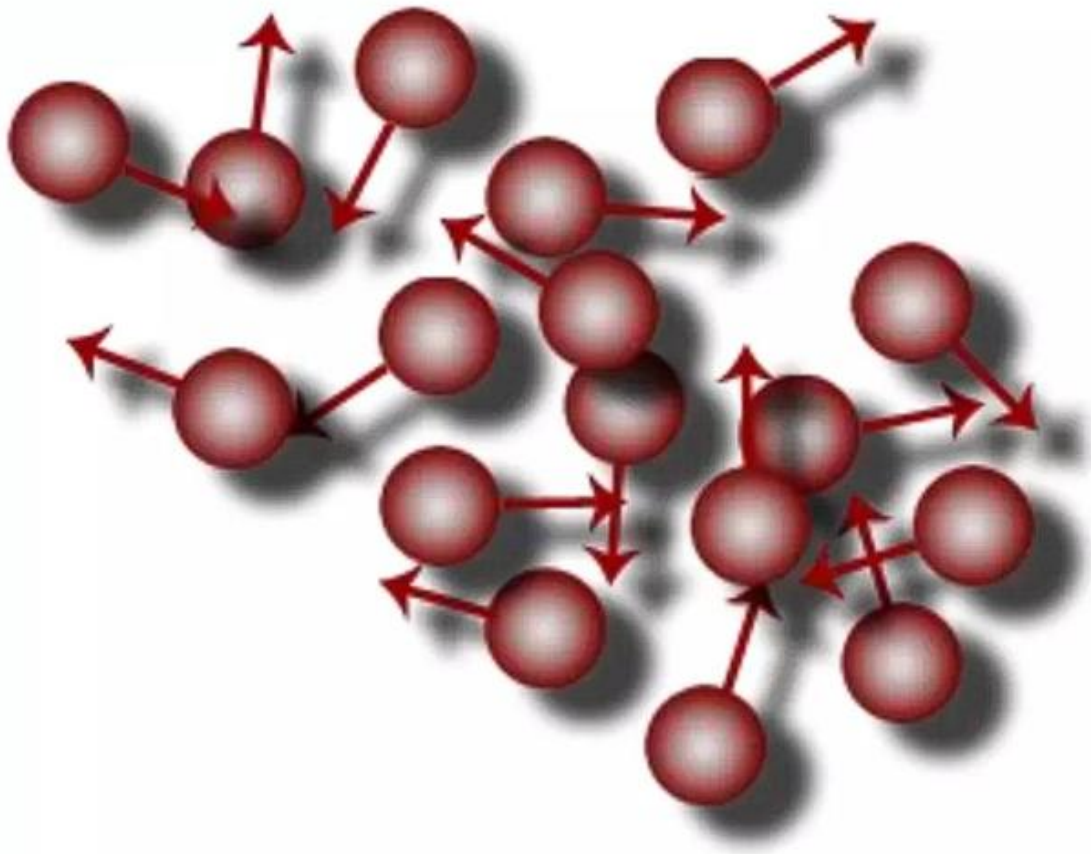
Gas Phase



Molecules move randomly with three different types of motions are vibration, rotation and translation.

Molecules are separated by large distances and travel a long way between collisions.

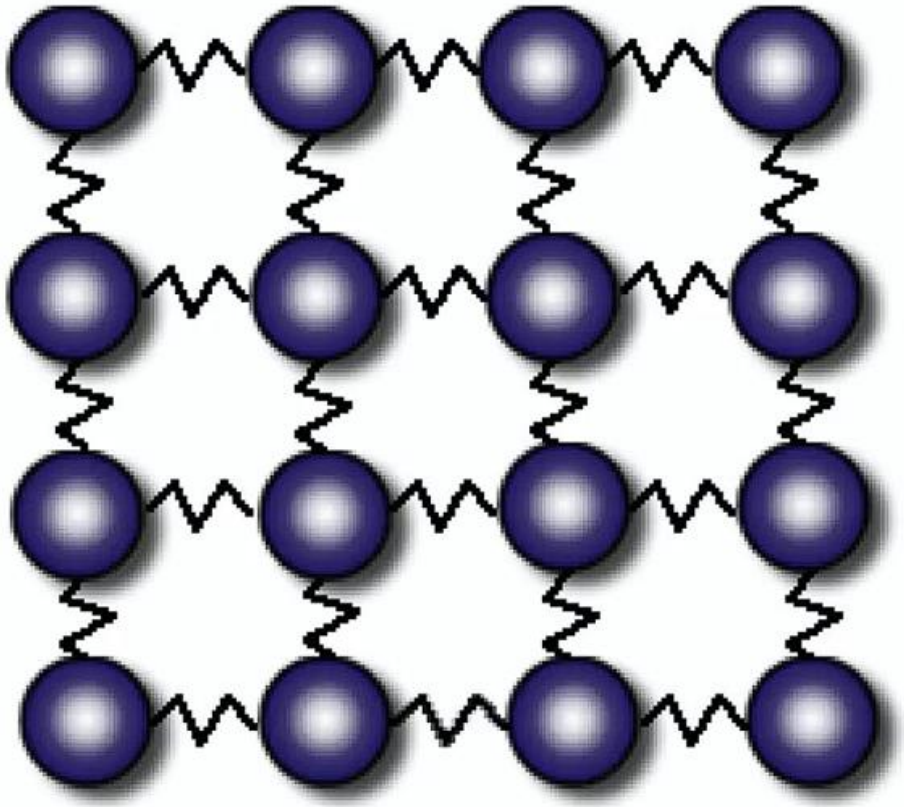
Liquid Phase



Molecules move randomly with all three types of motions, are vibration, rotation and translation.

But they are much closer together and cannot travel very far between collisions.

Solid Phase



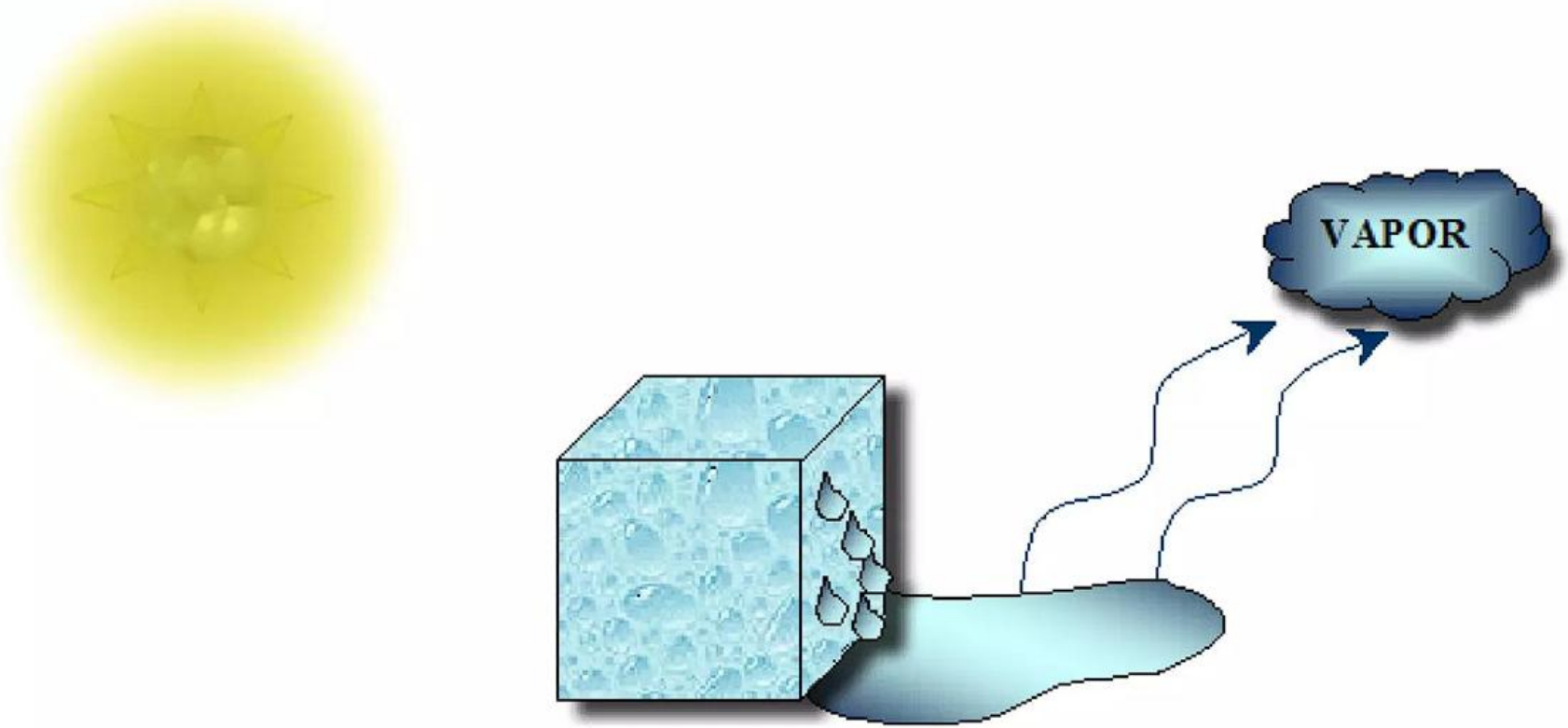
Atoms or molecules have all three types of motion, but they are very close together. As a result, they cannot travel far at all before they collide.

Each molecule moves about within a small space and does not tend to wander.

In a solid, the attractive and repulsive forces between the molecules tend to maintain them at relatively constant distances from each other.

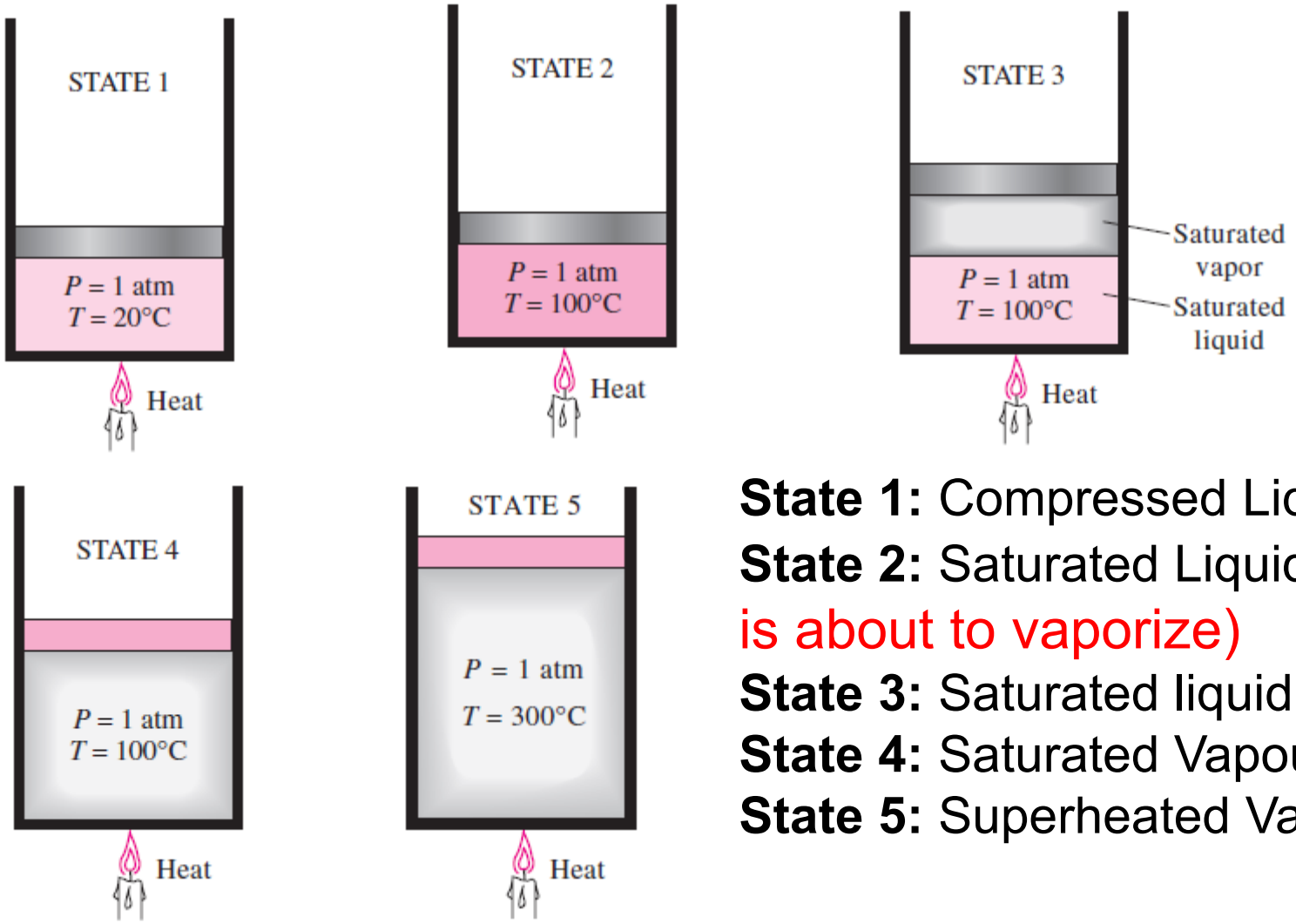
Phase Change Processes

Principle Phases: Solid (ice), Liquid (Water), and Gas (Water Vapor).



- Sunlight causes water molecules to leave the solid phase and enter the liquid phase.
- Sunlight also causes water molecules to leave the liquid phase and enter the gas phase.

Phase Change Processes in a Pure Substance



State 1: Compressed Liquid
State 2: Saturated Liquid (A liquid that is about to vaporize)
State 3: Saturated liquid + Vapour
State 4: Saturated Vapour
State 5: Superheated Vapour

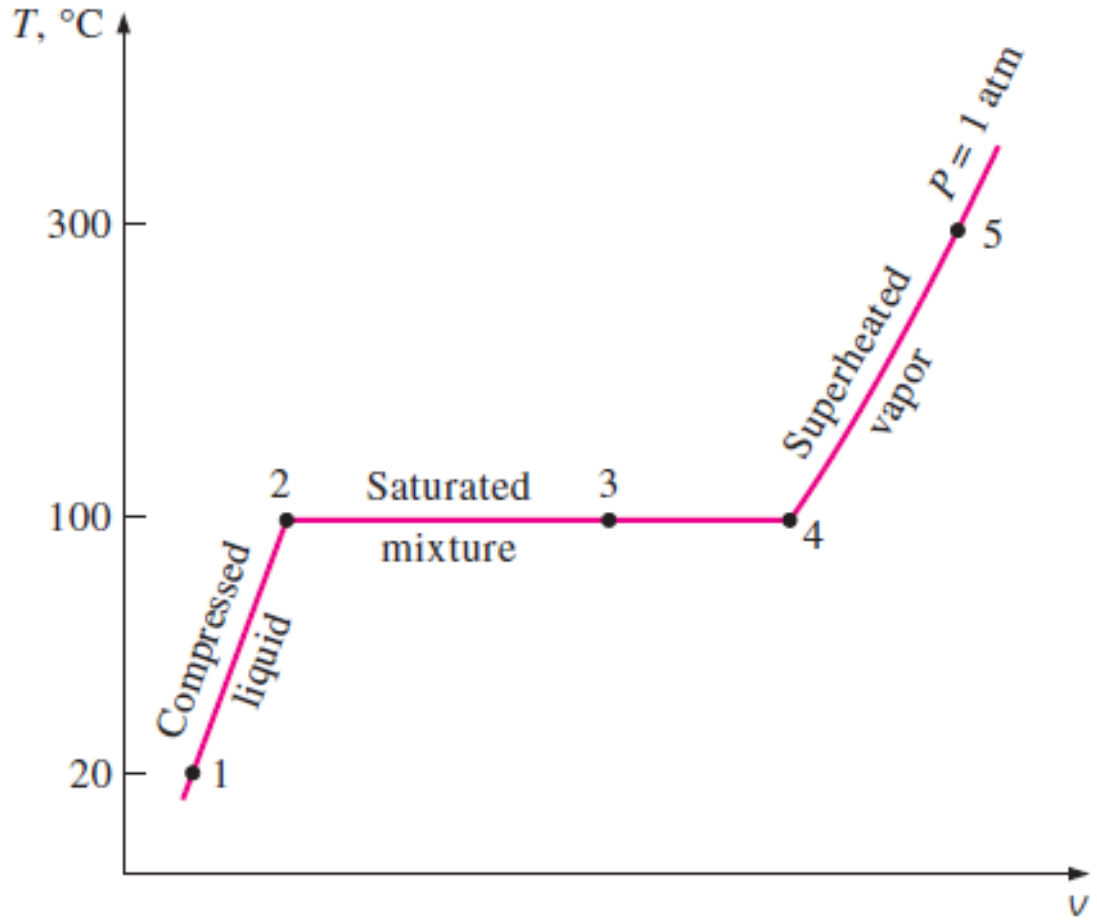
Definitions

- **Compressed or Sub-Cooled** liquid means that it is not about to vaporize
- **Saturated Liquid** is a liquid that is about to vaporize
- **Saturated Vapour** is a vapour that is about to condense
- **Saturated Liquid-Vapour Mixture** is a condition where liquid and vapour coexist in equilibrium
- **Superheated Vapour** is a vapour that is not about to condense

Definitions…

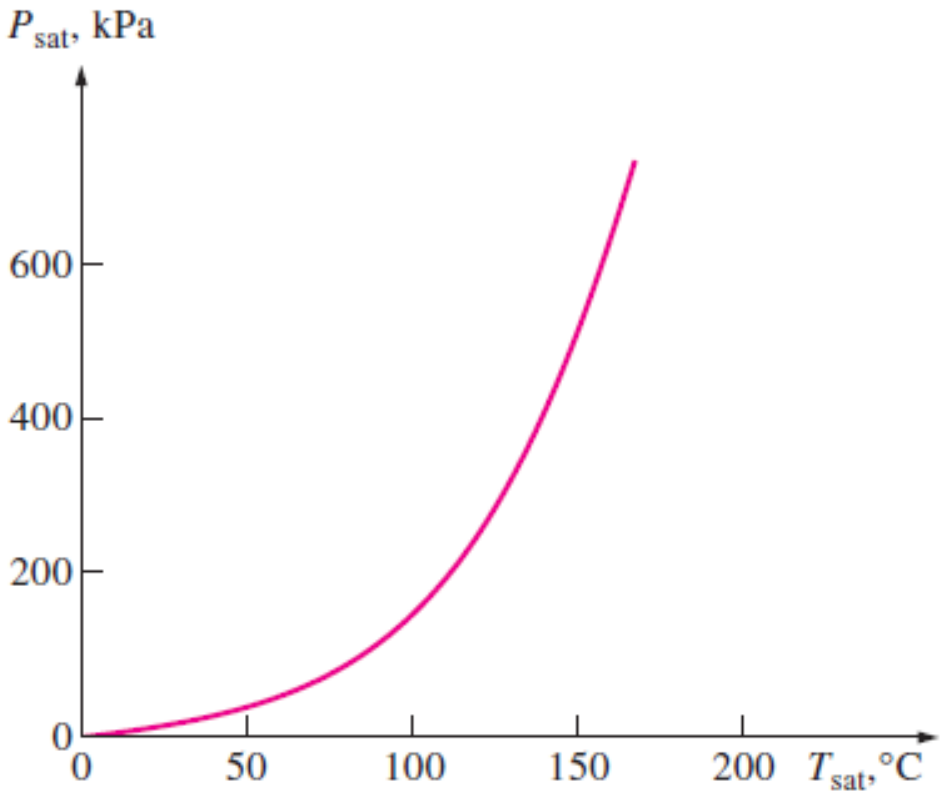
- **Saturation Temperature (T_{sat})** means the temperature at which change of phase takes place at a given pressure.
- **Saturation Pressure (P_{sat})** means the pressure at which change of phase takes place at a given temperature.
- **Latent heat:** the amount of energy absorbed or released during a phase-change process.
- **Latent heat of fusion:** the amount of energy absorbed during melting.
- **Latent Heat of Vaporization** is the amount of energy absorbed during evaporation
- **Sensible Heat** is the energy required to change the temperature of a substance with no phase change

Definitions…



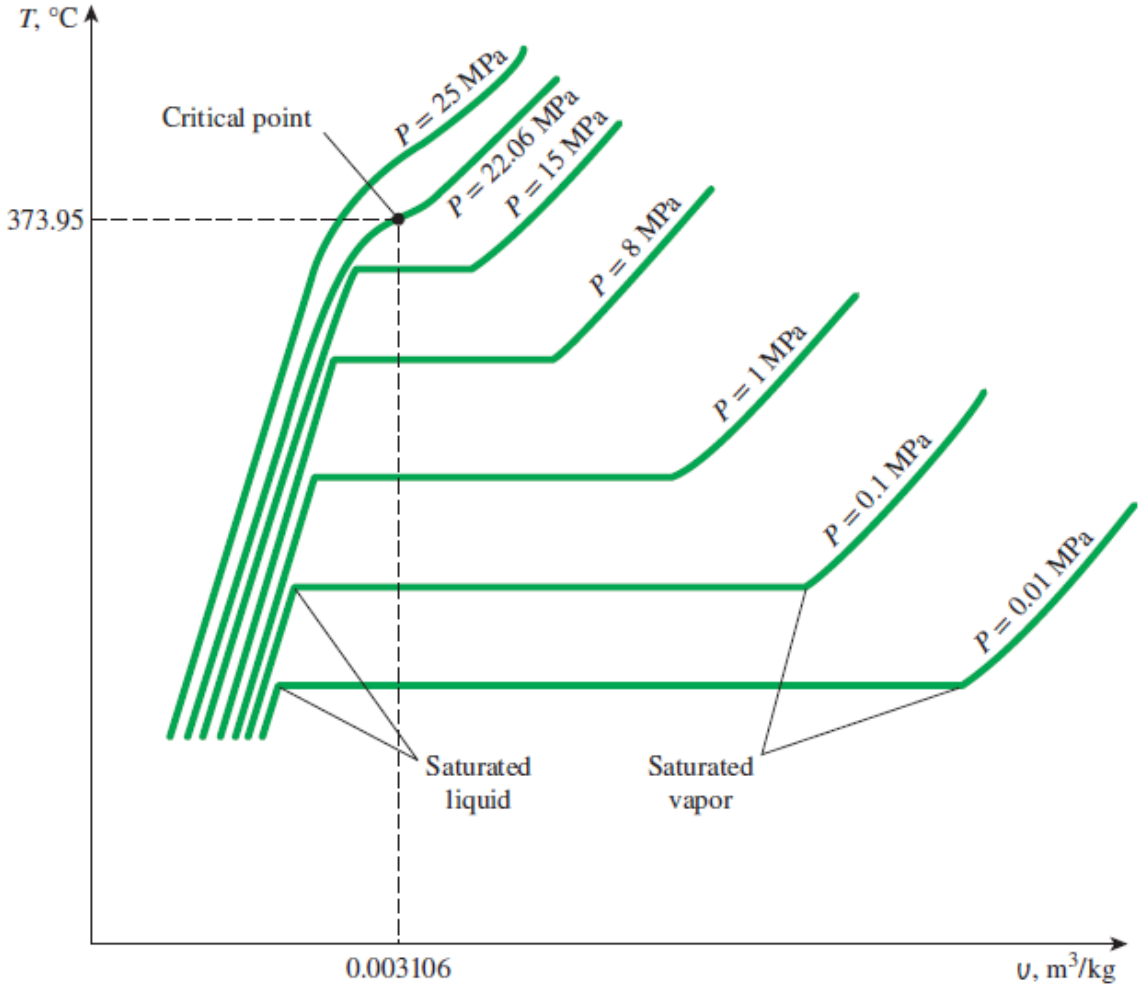
Liquid-Vapour Saturation Curve

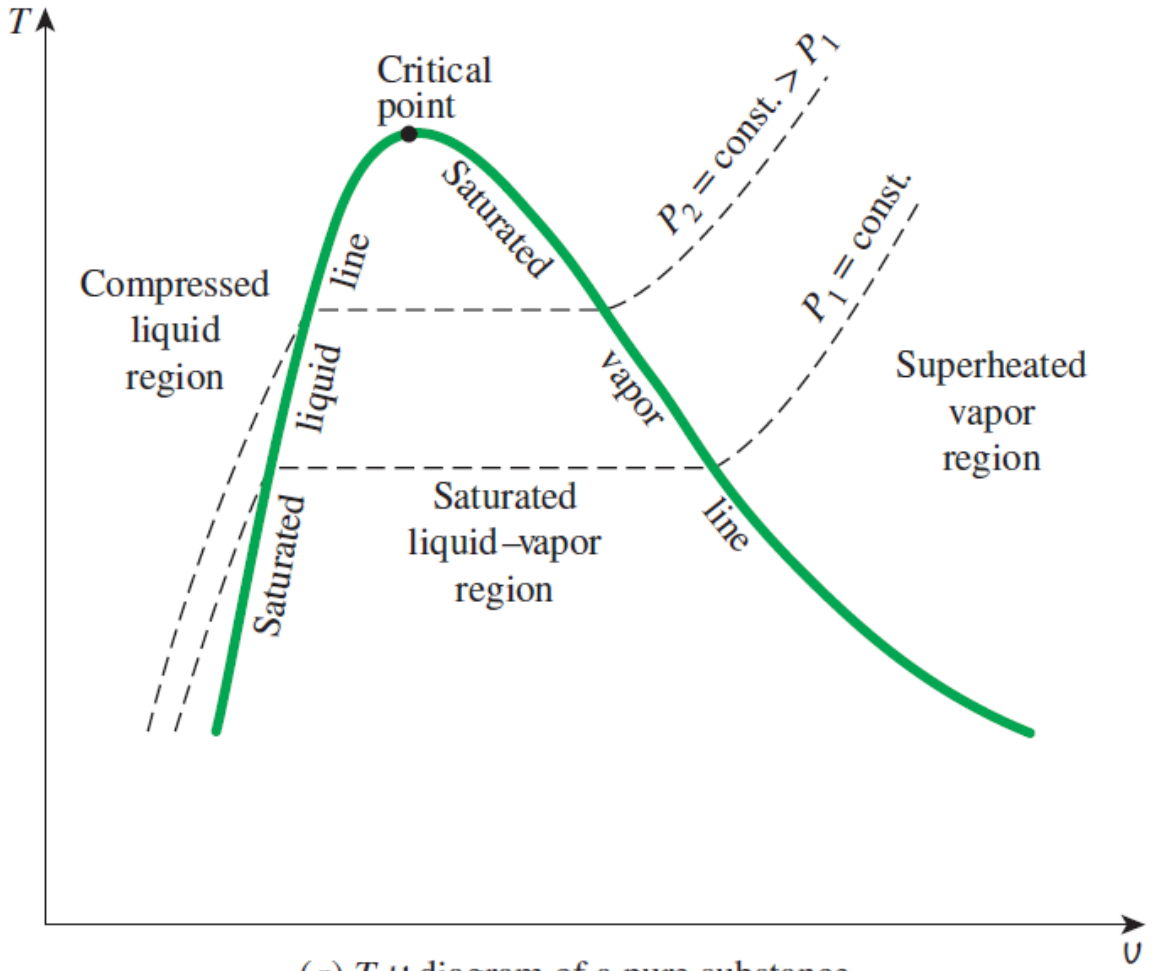
$T_{sat}=f(P_{sat})$



Property Diagrams for Phase-Change Processes

1. T-V Diagram

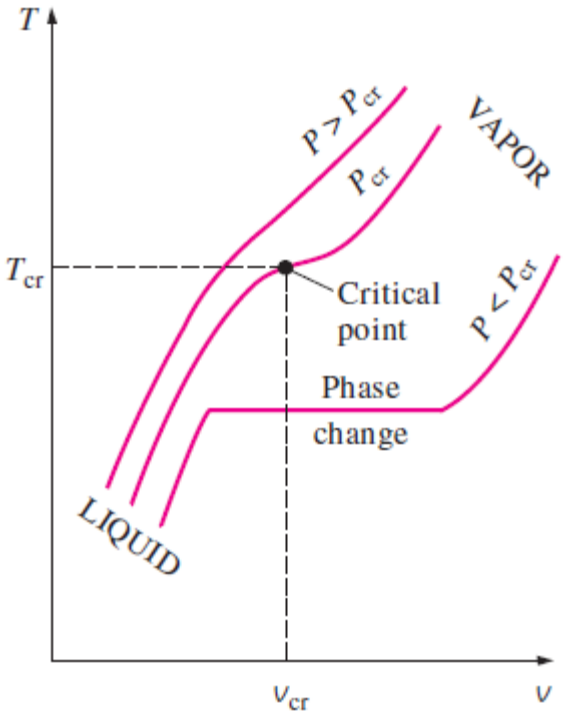




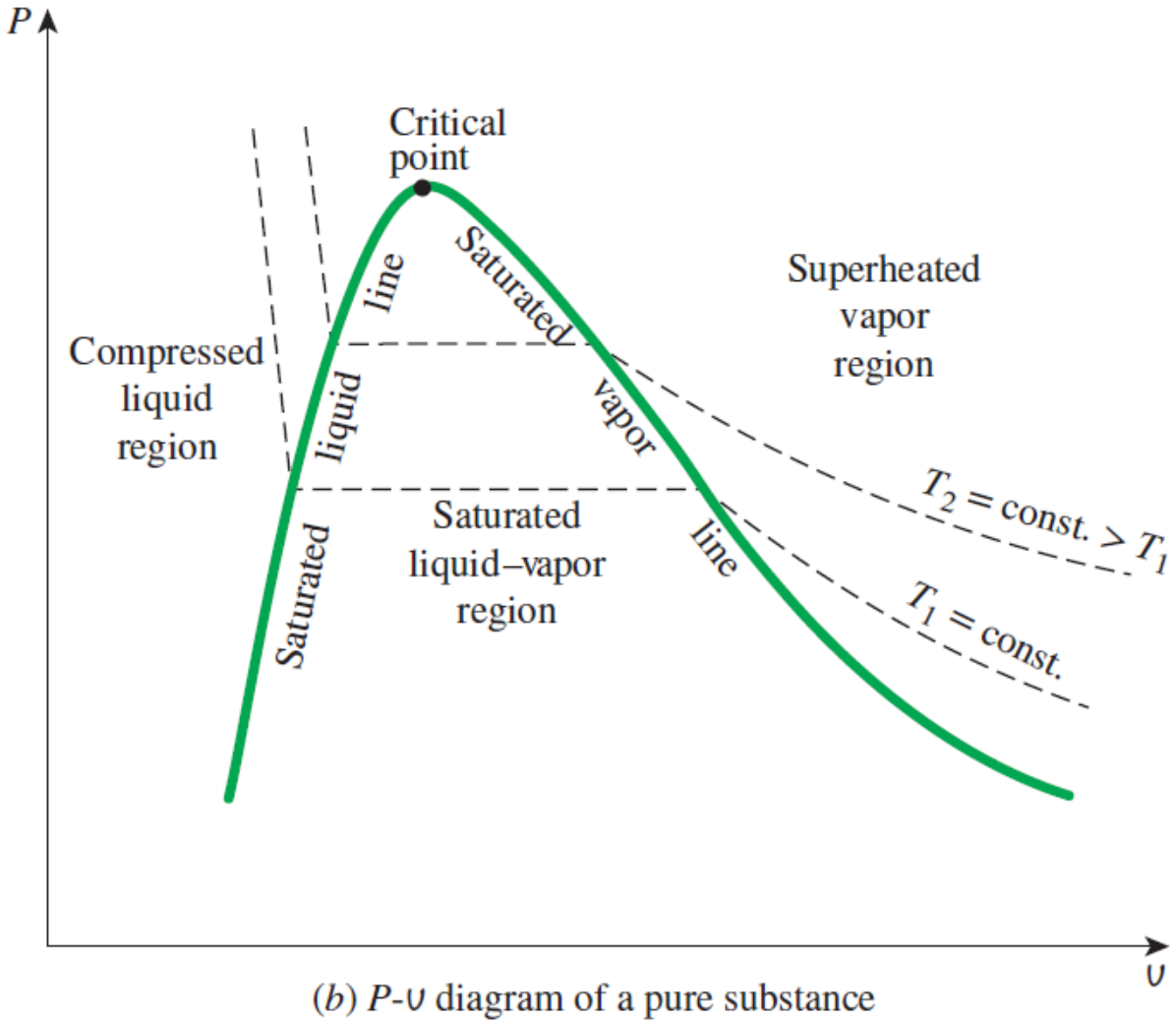
(*a*) *T*-υ diagram of a pure substance

Critical Point

- Critical Point is defined as the point at which the saturated liquid and saturated vapour states are identical. Or two lines meet at this point.
- For water, P_{cr}=22.06 Mpa
- T_{cr}=373.95 ^{0}C



2. P-V Diagram



(*b*) *P*-υ diagram of a pure substance

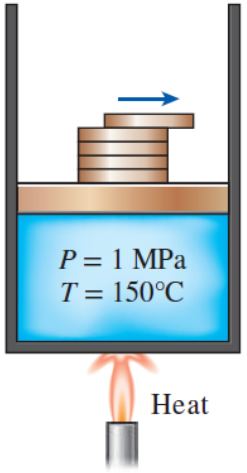


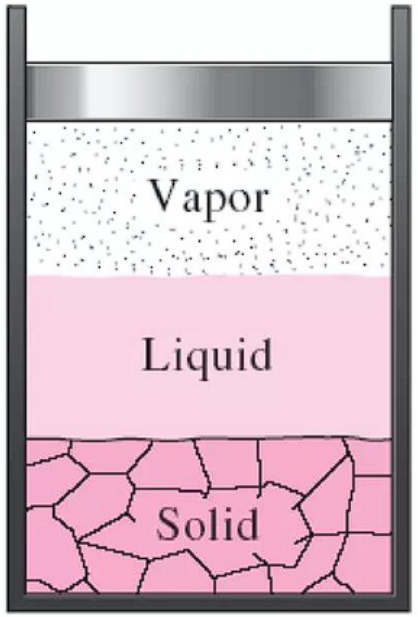
FIGURE 3–18
The pressure in a piston–cylinder device can be reduced by reducing the weight of the piston.

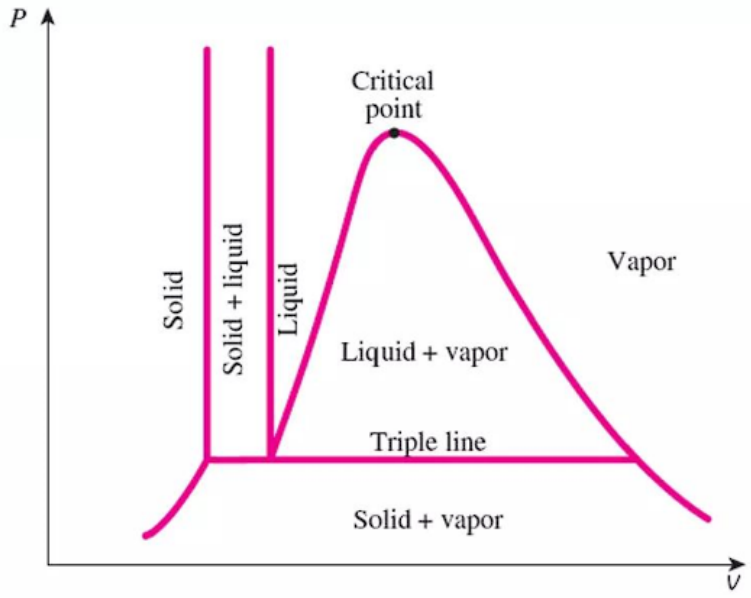
- The general shape of the *P-v* diagram of a pure substance is very much like the *T-v* diagram, but the *T* = constant lines on this diagram have a downward trend.

2. P-V Diagram extended to solid phase

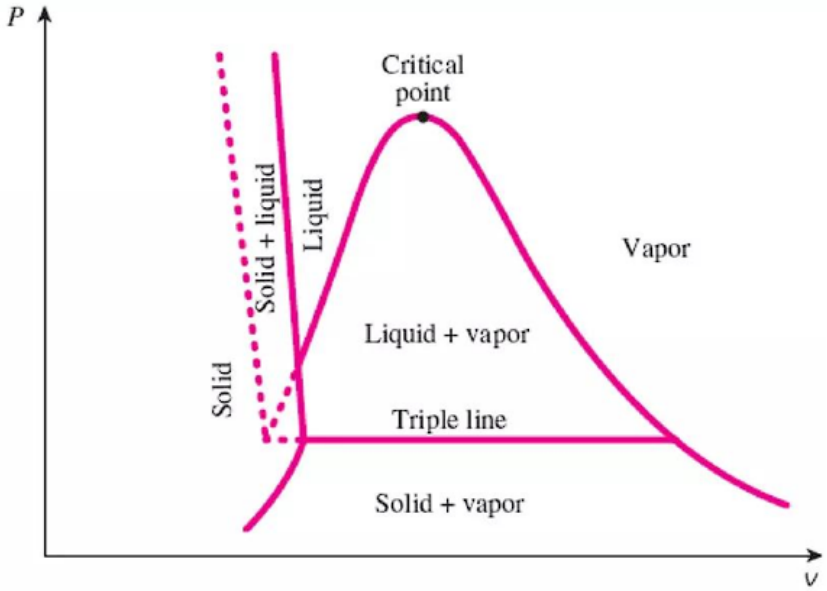
The variations of properties during phase-change processes are best studied and understood with the help of property diagrams such as the *P-v* diagrams for pure substances.

For water,
$T_{tp} = 0.01°C$
$P_{tp} = 0.6117$ kPa





(*a*) *P-v* diagram of a substance that contracts on freezing



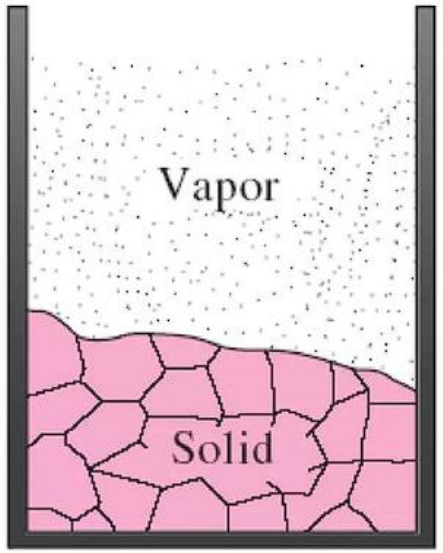
(*b*) *P-v* diagram of a substance that expands on freezing (such as water)

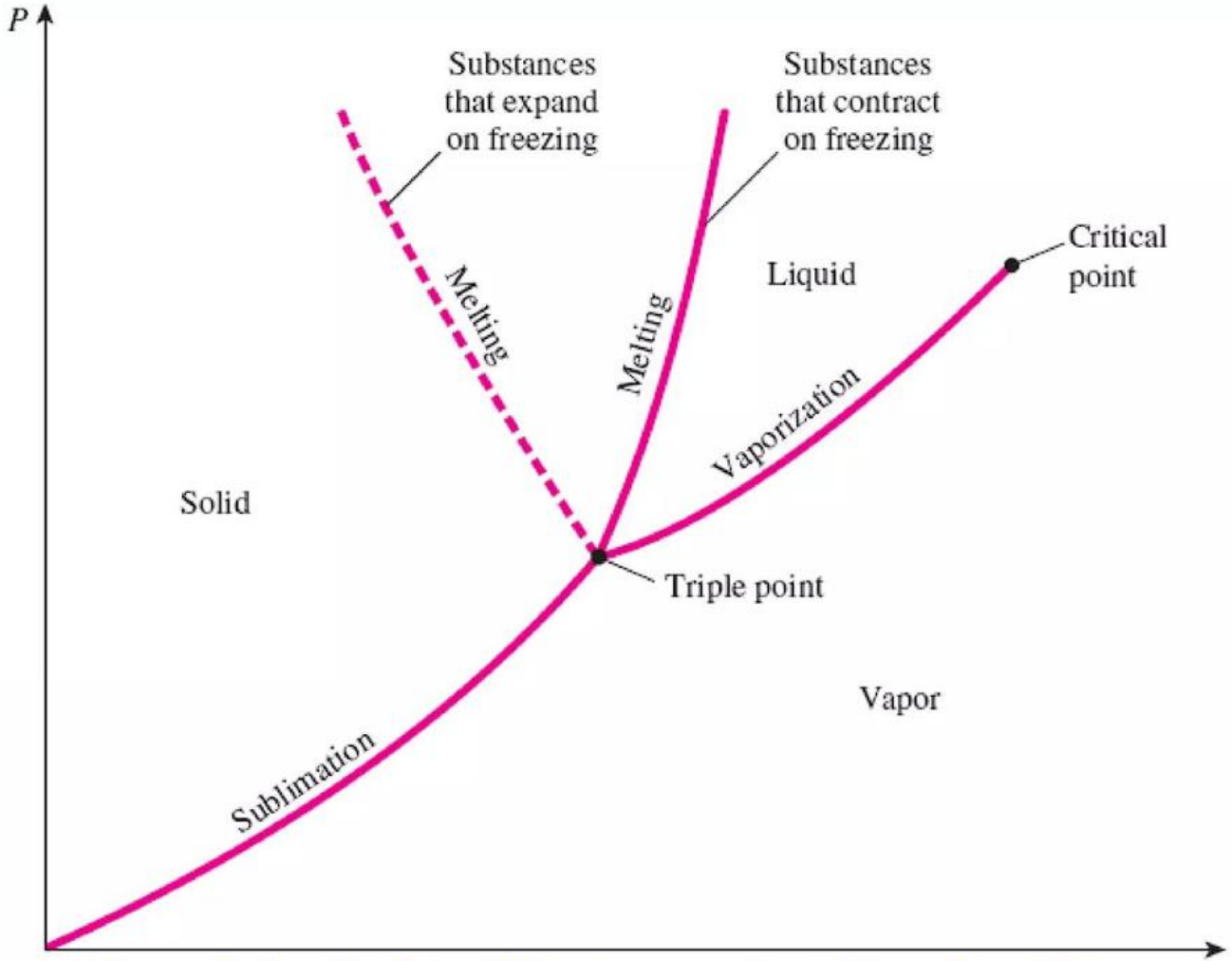
At triple-point pressure and temperature, a substance exists in three phases in equilibrium.

3. P-T Diagram

The variations of properties during phase-change processes are best studied and understood with the help of property diagrams such as the *P*-*T* diagrams for pure substances.

Sublimation: Passing from the solid phase directly into the vapor phase.





At low pressures (below the triple-point value), solids evaporate without melting first (*sublimation*).

Vapour (Steam) Tables

Table I (A-4) **and** ***Table II (A-5) :***

Liquid – Vapour Saturation State.

Table I : Temp. **– based** **&** ***Table II : Pr.*** **– based**

v_f, v_g $\quad$ u_f, u_g $\quad$ h_f, h_g $\quad$ s_f, s_g

$u_{fg} = u_g - u_f$ $\quad$ $h_{fg} = h_g - h_f$ $\quad$ $s_{fg} = s_g - s_f$

Triple Point : $u_f = 0$ *kJ/kg.*

$S_f = 0$ *kJ/kg.K* } *Definition*

$T_{TP} = 0.01\ ^oC$

Critical Point : $u_f = u_g$

L – V diff. vanishes..!! $\quad$ $s_f = s_g$

$v_f = v_g$

Vapour (Steam) Tables

TABLE A–4

Saturated water—Temperature table

Temp., T °C	Sat. Press., P_{sat} kPa	*Specific volume, m³/kg* Sat. liquid, v_f	*Specific volume, m³/kg* Sat. vapor, v_g	*Internal energy, kJ/kg* Sat. liquid, u_f	*Internal energy, kJ/kg* Evap., u_{fg}	*Internal energy, kJ/kg* Sat. vapor, u_g	*Enthalpy, kJ/kg* Sat. liquid, h_f	*Enthalpy, kJ/kg* Evap., h_{fg}	*Enthalpy, kJ/kg* Sat. vapor, h_g	*Entropy, kJ/kg·K* Sat. liquid, s_f	*Entropy, kJ/kg·K* Evap., s_{fg}	*Entropy, kJ/kg·K* Sat. vapor, s_g
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.001	2500.9	2500.9	0.0000	9.1556	9.1556
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21.020	2489.1	2510.1	0.0763	8.9487	9.0249
10	1.2281	0.001000	106.32	42.020	2346.6	2388.7	42.022	2477.2	2519.2	0.1511	8.7488	8.8999
15	1.7057	0.001001	77.885	62.980	2332.5	2395.5	62.982	2465.4	2528.3	0.2245	8.5559	8.7803
20	2.3392	0.001002	57.762	83.913	2318.4	2402.3	83.915	2453.5	2537.4	0.2965	8.3696	8.6661

TABLE A–5

Saturated water—Pressure table

Press., P kPa	Sat. temp., T_{sat} °C	*Specific volume, m³/kg* Sat. liquid, v_f	*Specific volume, m³/kg* Sat. vapor, v_g	*Internal energy, kJ/kg* Sat. liquid, u_f	*Internal energy, kJ/kg* Evap., u_{fg}	*Internal energy, kJ/kg* Sat. vapor, u_g	*Enthalpy, kJ/kg* Sat. liquid, h_f	*Enthalpy, kJ/kg* Evap., h_{fg}	*Enthalpy, kJ/kg* Sat. vapor, h_g	*Entropy, kJ/kg·K* Sat. liquid, s_f	*Entropy, kJ/kg·K* Evap., s_{fg}	*Entropy, kJ/kg·K* Sat. vapor, s_g
1.0	6.97	0.001000	129.19	29.302	2355.2	2384.5	29.303	2484.4	2513.7	0.1059	8.8690	8.9749
1.5	13.02	0.001001	87.964	54.686	2338.1	2392.8	54.688	2470.1	2524.7	0.1956	8.6314	8.8270
2.0	17.50	0.001001	66.990	73.431	2325.5	2398.9	73.433	2459.5	2532.9	0.2606	8.4621	8.7227
2.5	21.08	0.001002	54.242	88.422	2315.4	2403.8	88.424	2451.0	2539.4	0.3118	8.3302	8.6421
3.0	24.08	0.001003	45.654	100.98	2306.9	2407.9	100.98	2443.9	2544.8	0.3543	8.2222	8.5765

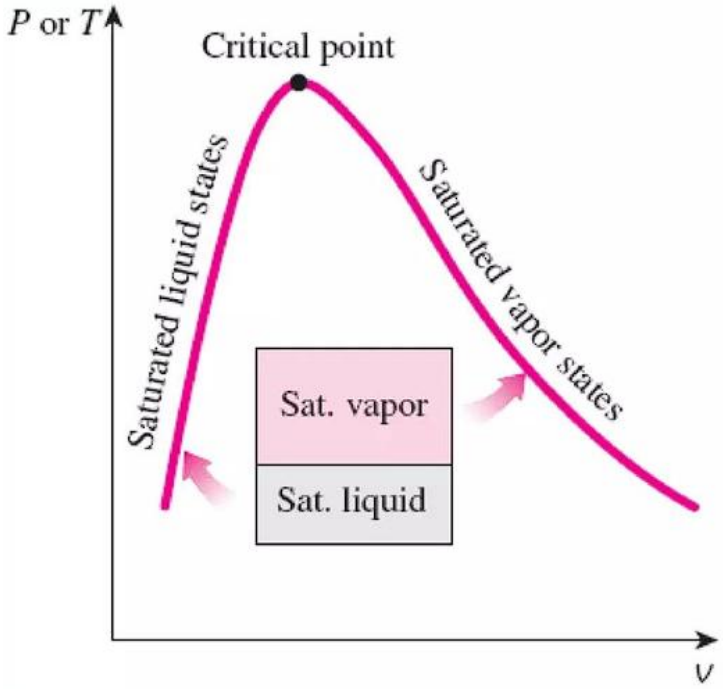
Liquid-Vapour Mixture (V Imp)

Quality, x : The ratio of the mass of vapor to the total mass of the mixture. **Quality is between 0 and 1 → 0: sat. liquid, 1: sat. vapor.**

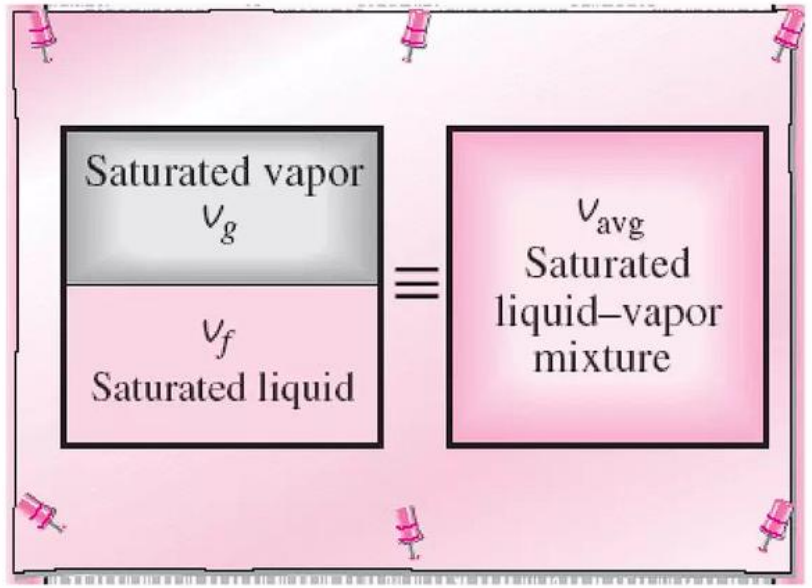
The properties of the saturated liquid are the same whether it exists alone or in a mixture with saturated vapor.

$$x = \frac{m_{\text{vapor}}}{m_{\text{total}}} \qquad m_{\text{total}} = m_{\text{liquid}} + m_{\text{vapor}} = m_f + m_g$$

Temperature and pressure are dependent properties for a mixture.



The relative amounts of liquid and vapor phases in a saturated mixture are specified by the ***quality x***.



A two-phase system can be treated as a homogeneous mixture for convenience.

Properties of Liquid-Vapour Mixture

Property	Original Equation	Simplified Form
Specific Volume (v)	$v = (1-x)v_f + xv_g$	$v = v_f + xv_{fg}$, where $v_{fg} = v_g - v_f$
Specific Internal Energy (u)	$u = (1-x)u_f + xu_g$	$u = u_f + xu_{fg}$, where $u_{fg} = u_g - u_f$
Specific Enthalpy (h)	$h = (1-x)h_f + xh_g$	$h = h_f + xh_{fg}$, where $h_{fg} = h_g - h_f$
Specific Entropy (s)	$s = (1-x)s_f + xs_g$	$s = s_f + xs_{fg}$, where $s_{fg} = s_g - s_f$
General Property (y)	$y = (1-x)y_f + xy_g$	$y = y_f + xy_{fg}$, where $y_{fg} = y_g - y_f$

- ***f*** is for ***Fluid/Liquid phase*** and ***g*** is for ***Gas/Vapor phase***

Characteristics of Superheated vapour

- Compared to saturated vapour, superheated vapour is characterized by

$$\text{Lower Pressures}\left(P < P_{sat} \text{ at a given } T\right)$$
$$\text{Higher Temperatures}\left(T > T_{sat} \text{ at a given } P\right)$$
$$\text{Higher Specific Volumes}\left(v > v_g \text{ at a given } P \text{ or } T\right)$$
$$\text{Higher internal energies}\left(u > u_g \text{ at a given } P \text{ or } T\right)$$
$$\text{Higher Enthalpies}\left(h > h_g \text{ at a given } P \text{ or } T\right)$$

Characteristics of Compressed Liquid

- A compressed liquid is characterized by

$$\text{Higher Pressures} \left(P > P_{sat} \text{ at a given } T\right)$$
$$\text{Lower Temperatures} \left(T < T_{sat} \text{ at a given } P\right)$$
$$\text{Lower Specific Volumes} \left(v < v_f \text{ at a given } P \text{ or } T\right)$$
$$\text{Lower internal energies} \left(u < u_f \text{ at a given } P \text{ or } T\right)$$
$$\text{Lower Enthalpies} \left(h < h_f \text{ at a given } P \text{ or } T\right)$$

- However, the properties of compressed liquid are not much different from the corresponding saturation liquid values

Interpolation

Formula

$$y = y_1 + (x - x_1)\frac{(y_2 - y_1)}{(x_2 - x_1)}$$

y = linear interpolation value

x = independent variable

x_1, y_1 = values of the function at one point

x_2, y_2 = values of the function at another point

EXAMPLE 3–1 Pressure of Saturated Liquid in a Tank

A rigid tank contains 50 kg of saturated liquid water at 90° C. Determine the pressure in the tank and the volume of the tank

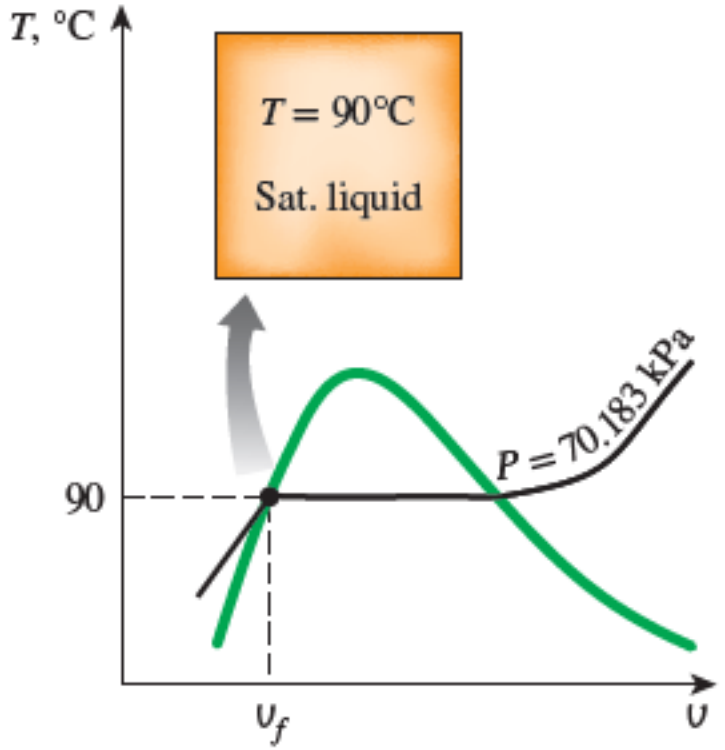


FIGURE 3–28
Schematic and *T*-*v* diagram for Example 3–1.

Problem - 1

Using steam tables, find the following at a pressure of 1.2 MPa: (i) saturation temperature; (ii) change in specific volume; (iii) change in entropy during evaporation; and (iv) latent heat of vaporisation of steam.

Solution - 1

$$\text{Saturation temperature, } t_{sat} = 187.99\ ^\circ\text{C}$$

$$\text{Specific volume of saturated liquid, } v_f = 0.001139\ \text{m}^3/\text{kg}$$

$$\text{Specific volume of saturated vapour, } v_g = 0.16333\ \text{m}^3/\text{kg}$$

$$\text{Change in specific volume} = v_g - v_f$$

$$= 0.16333 - 0.001139$$

$$= 0.162191\ \text{m}^3/\text{kg}$$

$$\text{Entropy of saturated liquid, } s_f = 2.2165$$

$$\text{Entropy of saturated vapour, } s_g = 6.5233$$

$$\text{Change in entropy during evaporation} = s_g - s_f$$

$$= 6.5233 - 2.2165 = 4.3068\ \text{kJ/kg K}$$

$$\text{Latent heat of vaporisation } h_{fg} = 1986.2\ \text{kJ/kg}$$

Problem - 2

Saturated steam has an entropy of 6.0018 kJ/kg K. What are its pressure, temperature, specific volume and enthalpy?

Solution - 2

From steam table, when $s_g = 6.0018$, $p = 4.6886$ MPa

$$t = 260\ ^\circ\text{C}$$

$$v_g = 0.04220\ \text{m}^3/\text{kg}$$

$$h_g = 2796.9\ \text{kJ/kg}$$

Problem - 3

A vessel of volume 0.05 m^3 contains a mixture of saturated water and saturated steam at temperature 250 °C. The mass of the liquid present is 10 kg. Find the pressure, mass, specific volume, enthalpy, entropy and internal energy.

Solution - 3

From steam table, at 250 °C, $p_{sat} = 3.973\ MPa$ and

$v_f = 0.0012512\ \text{m}^3/\text{kg}$ $h_f = 1085.36\ \text{kJ/kg}$ $s_f = 2.7927\ kJ/\ \text{kg K}$

$v_g = 0.05013\ \text{m}^3/\text{kg}$ $h_{fg} = 1716.2\ \text{kJ/kg}$ $s_{fg} = 3.2802\ \text{kJ/kg K}$

Volume of liquid, $V_f = mv_f = 10 \times 0.0012512 = 0.012512\ \text{m}^3$

Volume of vapour V_g = Volume of vessel − Volume of liquid

$V_g = 0.05 - 0.012512 = 0.037488\ \text{m}^3$

Mass of vapour, $m_g = \dfrac{V_g}{v_g} = \dfrac{0.037488}{0.05013} = 0.74782\ \text{kg}$

Total mass of mixture, $m = m_f + m_g = 10 + 0.74782 = 10.74782\ \text{kg}$

Quality of mixture $x = \dfrac{m_g}{m} = \dfrac{0.74782}{10.74782} = 0.06958$

$$v = v_f + x v_{fg} = 0.012512 + 0.06958 \times (0.05013 - 0.012512) = 0.0151 \text{ m}^3/\text{kg}$$

$$h = h_f + x h_{fg} = 1085.36 + 0.074782 \times 1716.2 = 1213.7 \text{ kJ/kg}$$

$$s = s_f + x s_{fg} = 2.7927 + 0.06958 \times 3.2802 = 3.0209 \text{ kJ/kg K}$$

$$u = h - pv = 1213.7 - 3.973 \times 10^3 \times 0.0151 = 1153.71 \text{ kJ/kg}$$

Problem - 4

Find the specific volume, enthalpy and entropy of steam at 1.4 MPa and 300 °C.

Solution - 4

At 1.4 MPa, from steam take $t_{sat} = 195.07$ °C. Therefore, the state of the steam will be superheated steam. Now, the properties of the superheated steam can be seen from steam table. At 1.4 MPa and 300 °C,

$$\vartheta = 0.18233 \quad \frac{m^3}{kg}$$

$$h = 3040.9 \quad \frac{kJ}{kg}$$

$$s = 6.9553 \quad \frac{kJ}{kg.K}$$

Problem - 5

Find the enthalpy and entropy of steam when the pressure is 2 MPa and specific volume is 0.09 m^3/kg.

Solution - 5

In steam table for p is 2 MPa, v_f=0.001177 m^3/kg and v_g = 0.9963 m^3/kg. Since the given volume lies between v_f and v_g, the steam will be a mixture of liquid and vapour. The state will be within the vapour dome. When in the two-phase region, the composition of the mixture or its quality should be evaluated first. For this,

$$v = v_f + x v_{fg}$$

$$0.09 = 0.001177 + x(0.09963 - 0.001177)$$

$$x = 0.902 \quad \text{or} \quad 90.2$$

At 2 MPa, $h_f = 908.79$ kJ/kg $h_{fg} = 1890.7$ kJ/kg

$s_f = 2.4474$ kJ/kg K $s_{fg} = 3.8935$ kJ/kg K

$$h = h_f + x h_{fg} = 908.79 + 0.902 \times 1890.7 = 2614.20 \text{ kJ/kg}$$

$$s = s_f + x s_{fg} = 2.4474 + 0.902 \times 3.8935 = 5.9594 \text{ kJ/kg K}$$

Problem - 6

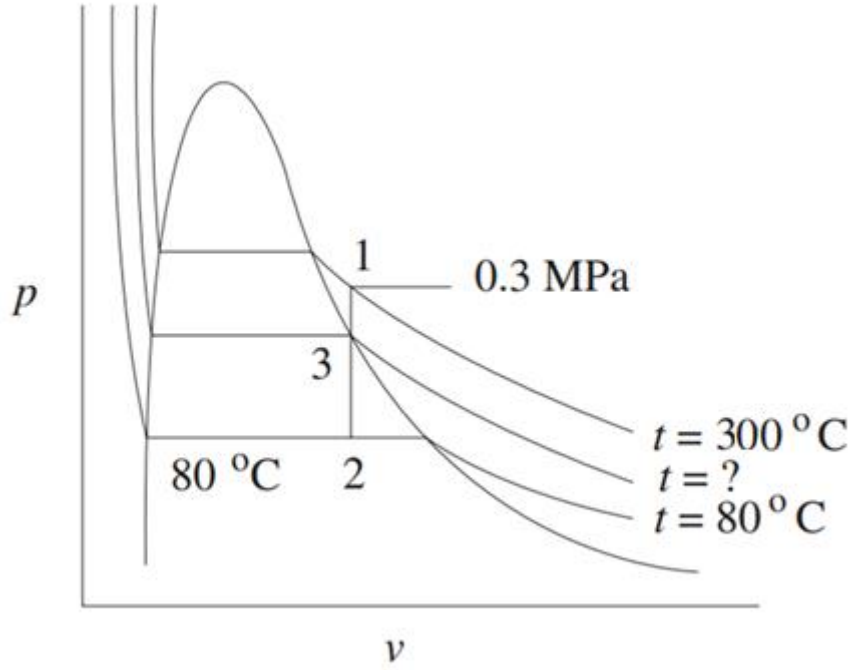
Consider a power plant in which steam is at a pressure of 0.3 MPa and temperature is 300 °C. It is cooled at constant volume.

(i) Find at what temperature, the steam becomes a saturated vapor.

(ii) What will be its quality at 80 °C?

(iii) What is the heat transfer per kg of steam by cooling it from 300 °C to 80 °C?

Solution - 6



Solution 6

Given:

- Initial pressure p=300kPa.
- Initial temperature, T_1=300°C=573 K

From the property table of water and steam, corresponding to the pressure of 0.3 MPa, we find the saturation temperature as T_{sat}=133.52°C.

Since, the initial temperature of the steam is T_1=300°C, the initial condition of the steam is superheated.

From the superheated table, find that the following properties,
Specific enthalpy h_1=3069.6kJ/kg.
specific volume v_1=0.87535m^3/kg

The change in state takes place at constant volume. So,
$v_1 = v_2$= 0.87535 m^3/kg.

We need to find the temperature corresponding to the specific volume of $v_1 = v_2$ = 0.87535m^3/kg.

Solution 6

From the saturation table, find the following properties.

$v_{120\circ C} = 0.89133\ m^3/kg$ and $v_{125\circ C} = 0.77102\ m^3/kg$.

Find the saturation temperature by interpolation.

$$T = T_1 + \frac{(v - v_1)}{(v_2 - v_1)} \cdot (T_2 - T_1)$$

$$T = 120 + \frac{(0.87535 - 0.89133)}{(0.77102 - 0.89133)} \cdot (125 - 120)$$

$$T = 120 + \frac{-0.01598}{-0.12031} \cdot 5$$

$$T \approx 120.66^\circ C$$

To find the quality of steam at 80°, find the following properties from the saturation table at at 80°.

$v_f = 0.001029\, m^3/kg.\ v_g = 3.4053\, m^3/kg.$

$$v_1 = v_2 = v_f + x_2(v_g - v_f)$$

$$0.87535 = 0.001029 + x_2(3.4053 - 0.001029)$$

$$\mathbf{x_2 = 0.257}$$

Solution 6

Find the enthalpy corresponding to 80°.

$$h_2 = h_{f2} + x_2 h_{fg2}$$
$$= 335.02 + 0.257(2308.0)$$
$$h_2 = 928.18\ kJ/kg$$

Change in enthalpy,

$$\Delta h = h_1 - h_2$$
$$= 3069.6 - 928.18$$
$$\mathbf{\Delta h = 2141.42\ kJ/kg}$$

Thus,

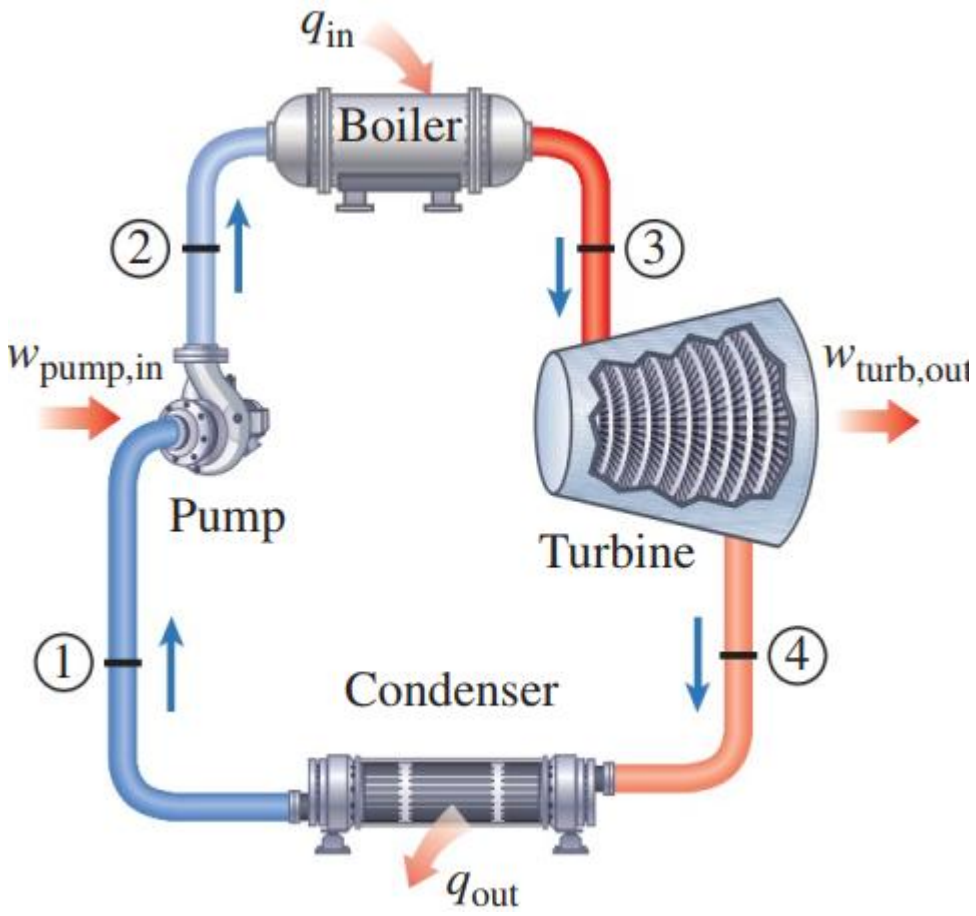
- (a) The saturation temperature $\mathbf{T_{sat} = 120.66^{\circ}.}$
- (b) The quality of steam at **80 ^{o}C is $\mathbf{x_2 = 0.257.}$**
- (c) The change in enthalpy during this process is $\mathbf{\Delta h = 2141.42\ kJ/kg}$

Ideal Rankine Cycle

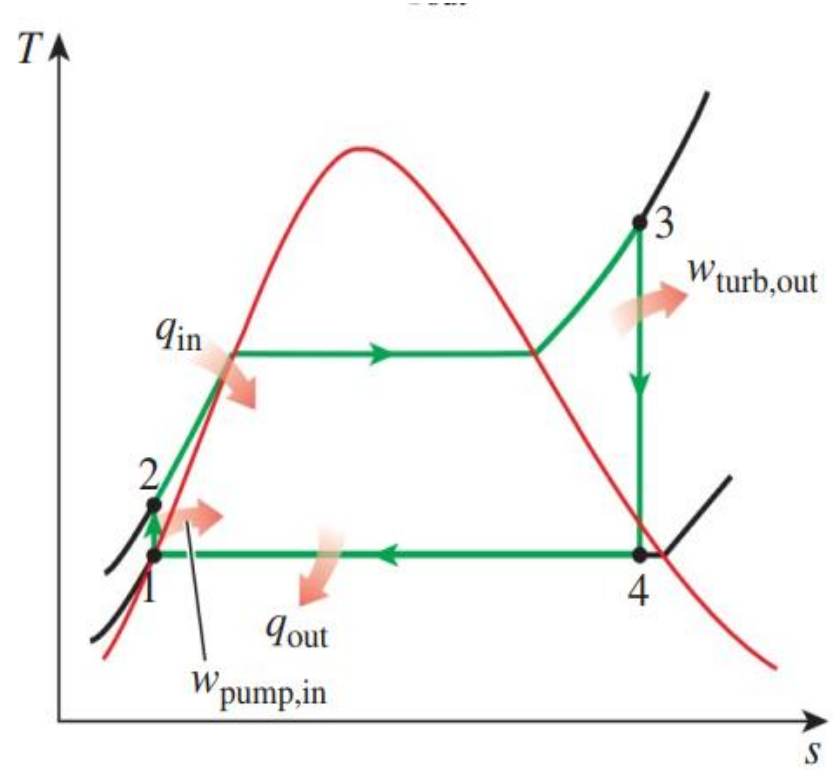
The processes in an Ideal Rankine Cycle are:

- Isentropic Compression in a Pump (1-2)
- Constant Pressure Heat Addition in a Boiler (2-3)
- Isentropic Expansion in a Turbine (3-4)
- Constant Pressure Heat Rejection in a Condenser (4-1)

Ideal Rankine Cycle…



Schematic Diagram



T-S Diagram

Energy Analysis of the Ideal Rankine Cycle

- The four devices operate under steady flow conditions
- KE and PE changes are negligible
- Boiler and Condenser do not involve any work
- Pump and Turbine operate Isentropically

Energy Analysis of the Ideal Rankine Cycle…

$$SFEE: Q - W = \Delta h$$

$$Pump\ (\mathrm{q} = 0): \quad w_{pump} = h_2 - h_1 = v(P_2 - P_1)$$

$$Boiler\ (\mathrm{w} = 0): \quad q_{in} = h_3 - h_2$$

$$Turbine\ (\mathrm{q} = 0): \quad w_{turbine} = h_3 - h_4$$

$$Condenser\ (\mathrm{w} = 0): \quad q_{out} = h_4 - h_1$$

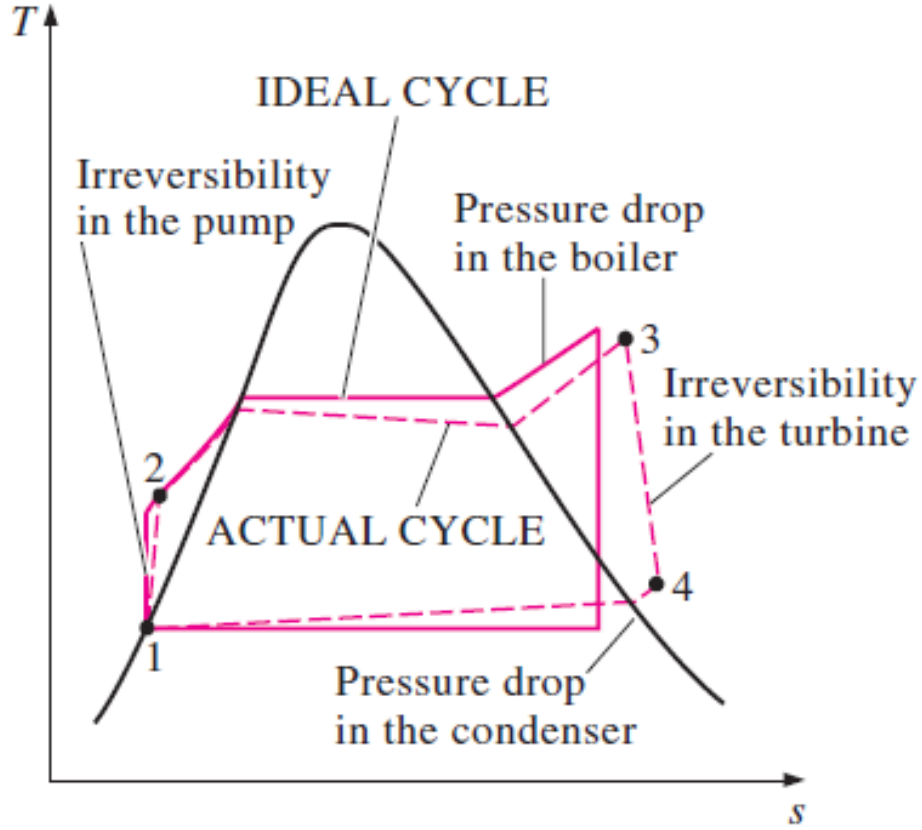
$$\eta_{th} = \frac{w_{net}}{q_{in}} = \frac{w_{turbine} - w_{pump}}{q_{in}} = 1 - \frac{q_{out}}{q_{in}}$$

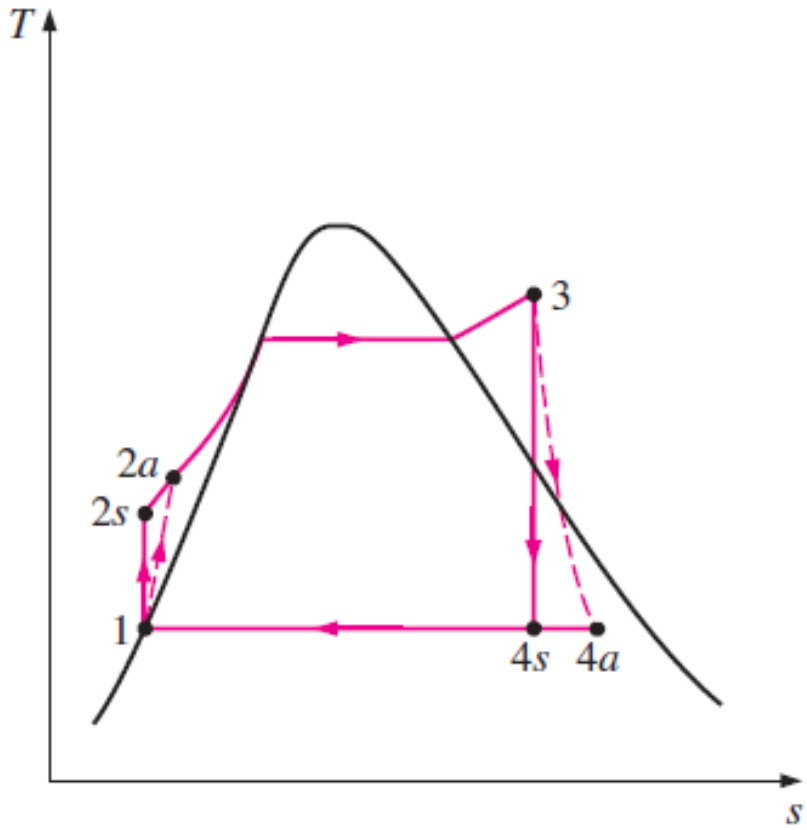
Heat Rate: Heat supplied in kJ to generate 1 kWh of electricity

Deviation of Actual Vapour Cycle from Ideal Cycle

- Deviations are a result of irreversibilities
- Fluid Friction causes pressure drop
- Heat loss from the steam to the surroundings results in lower efficiency
- Due to irreversibilities in pump and turbine, a pump requires more work and a turbine produces less work

Deviation of Actual Vapour Cycle from Ideal Cycle…





$$\eta_{pump} = \frac{w_{isentropic}}{w_{actual}} = \frac{h_{2s} - h_1}{h_{2a} - h_1};$$

$$\eta_{turbine} = \frac{w_{actual}}{w_{isentropic}} = \frac{h_3 - h_{4a}}{h_3 - h_{4s}}$$

Methods to Improve Efficiency of Rankine Cycle

Basic Principles used are:

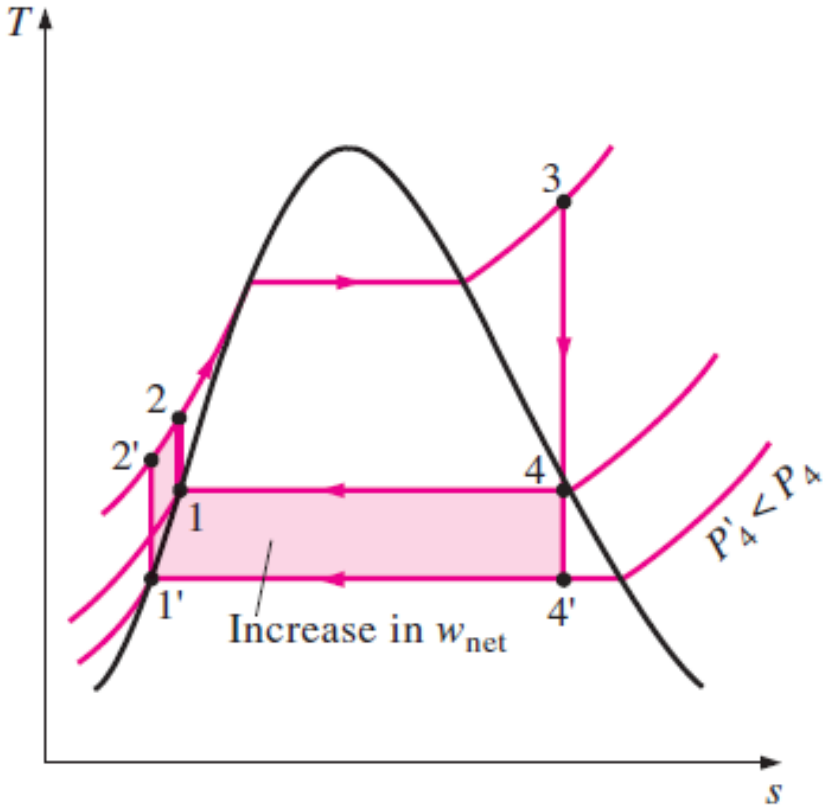
- Increase the average temperature of heat addition
- Decrease the average temperature of heat rejection

Methods Adopted are:

- Lowering the Condenser Pressure
- Superheating the steam to high temperatures
- Increasing the Boiler Pressure

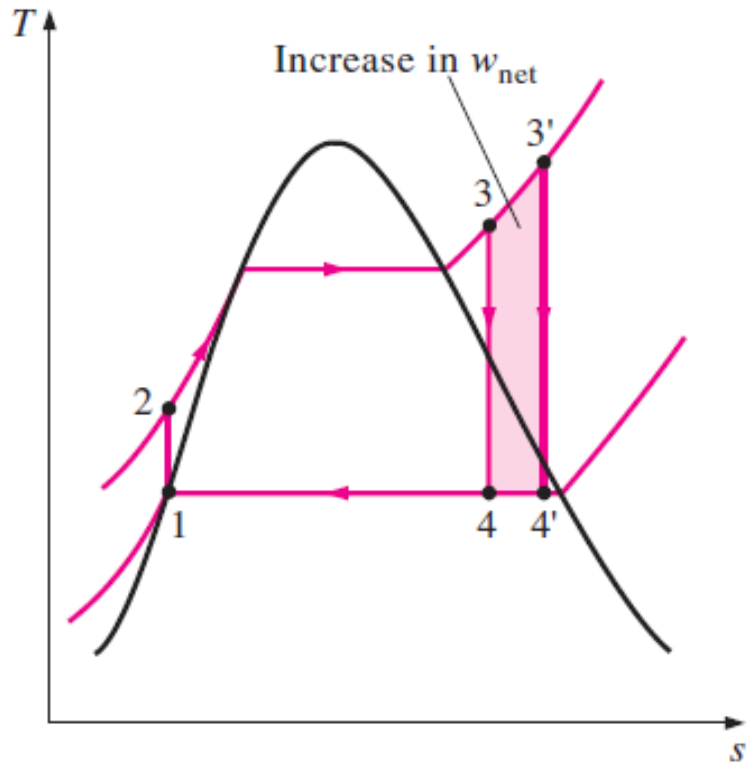
Methods to Improve Efficiency of Rankine Cycle

1. Lowering the Condenser Pressure (Lowers $T_{low,avg}$)



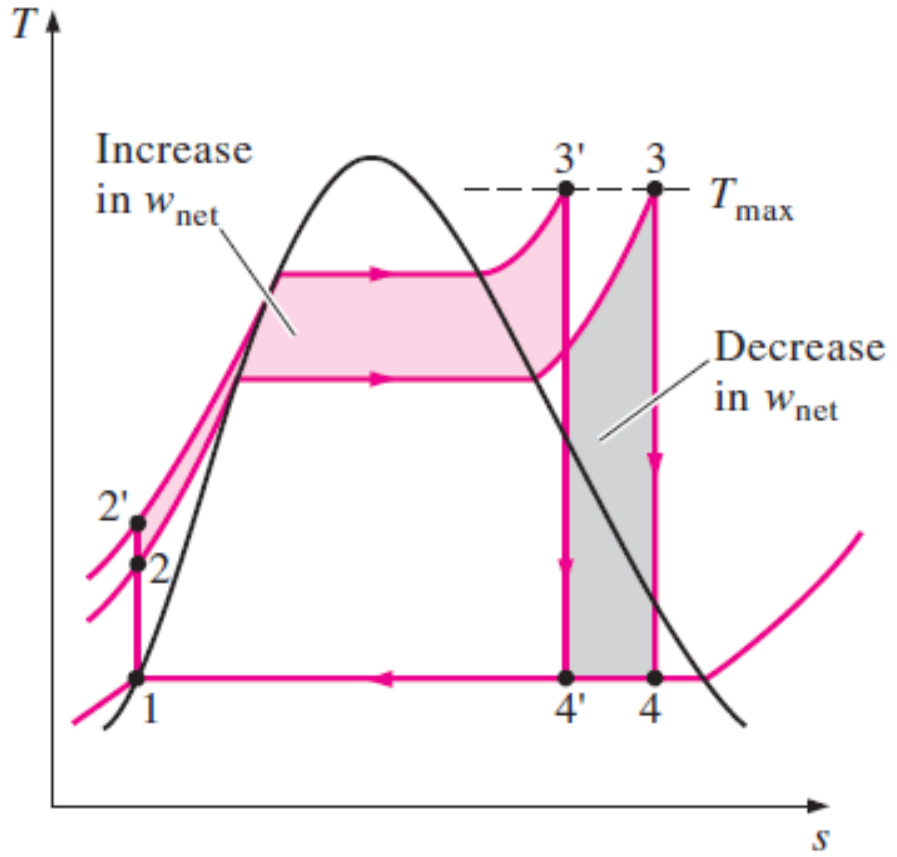
Methods to Improve Efficiency of Rankine Cycle

2. Superheating the steam to high temperatures (Increases $T_{high,avg}$)

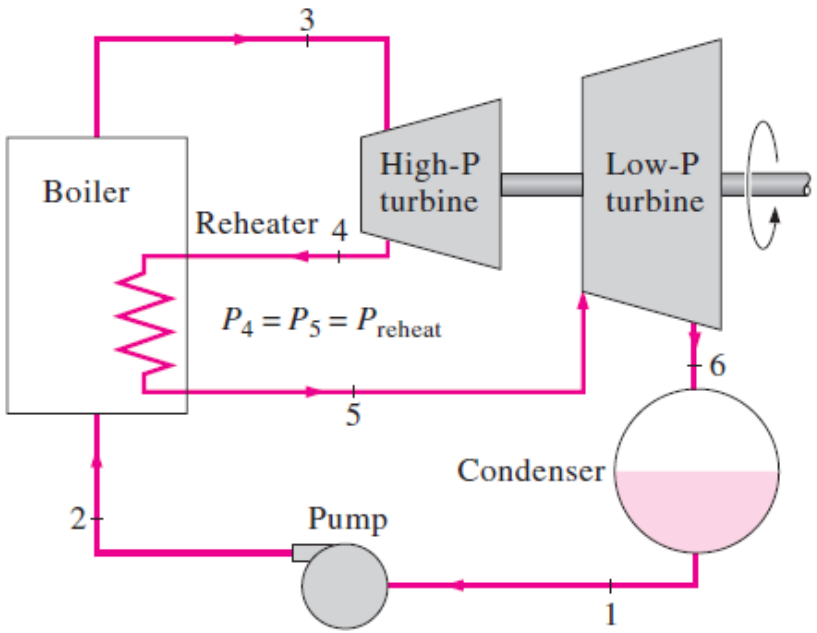


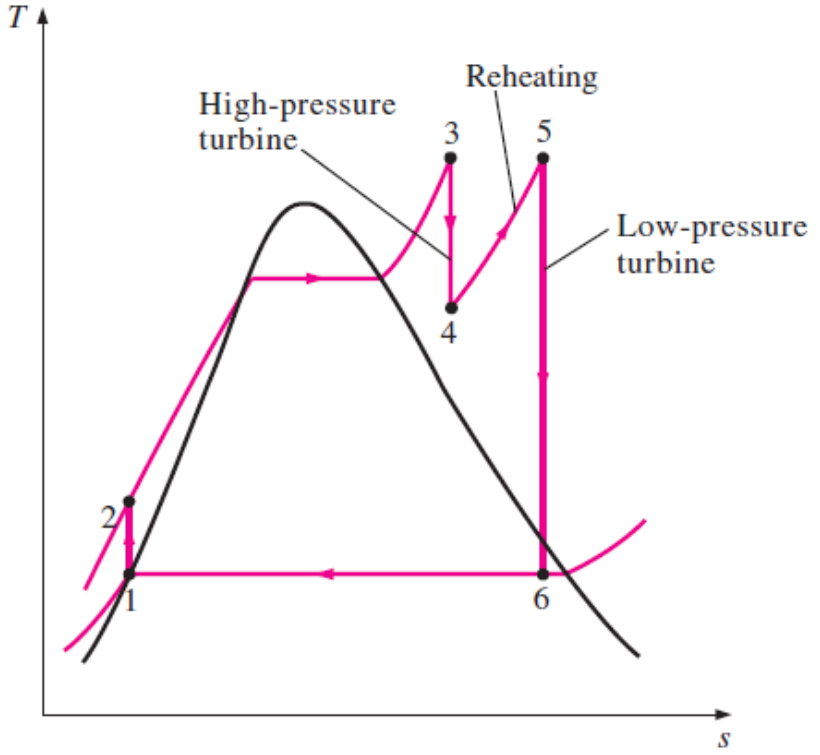
Methods to Improve Efficiency of Rankine Cycle

3. Increasing the Boiler Pressure (Increases $T_{high,avg}$)



Ideal Reheat Rankine Cycle

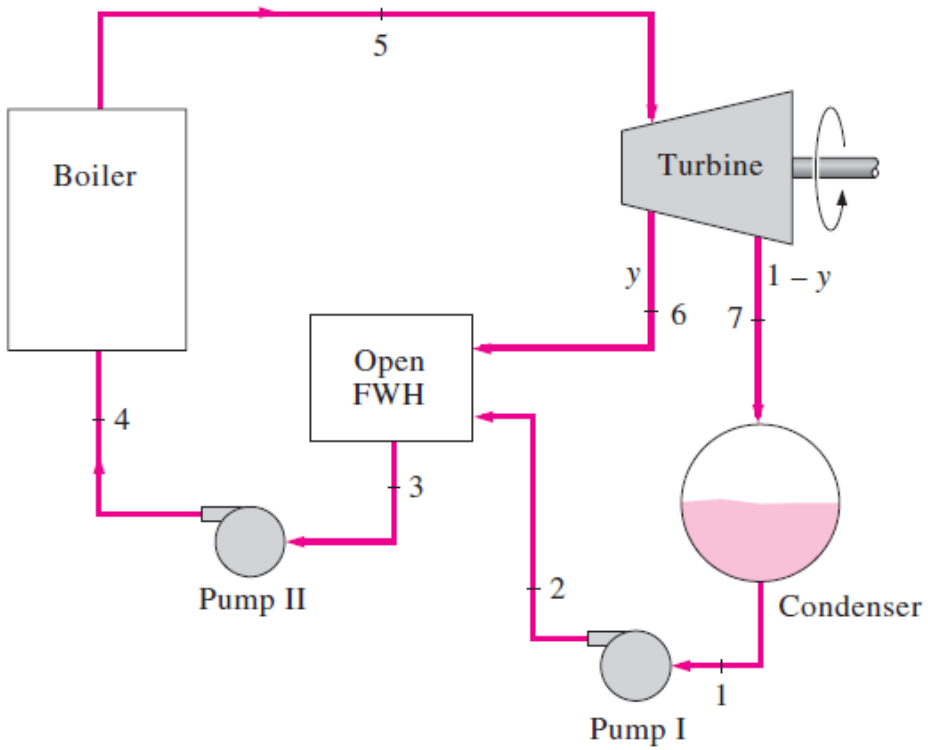


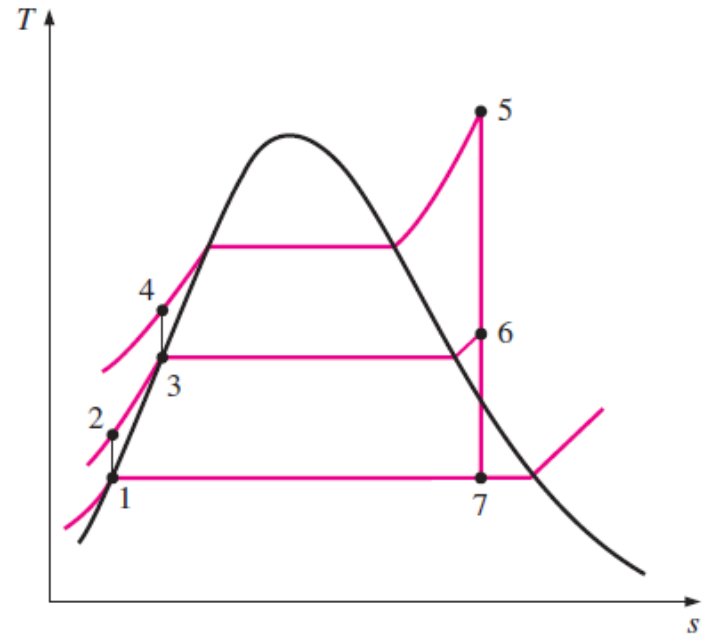


$$q_{in} = q_{primary} + q_{reheat} = (h_3 - h_2) + (h_5 - h_4)$$

$$w_{turbine} = w_{turbineI} + w_{turbineII} = (h_3 - h_4) + (h_5 - h_6)$$

Ideal Regenerative Rankine Cycle





Steam Rate:

- It is defines as the ratio of amount of steam (Kg) or steam flow in Kg/hr required to produce unit shaft power output (1KW).

$$SSC = \frac{\text{Mass of steam (m)in Kg/hr}}{\text{Power outputin KW}}$$

$$= \frac{m\ (kg/hr)}{m\ (kg/\sec) \text{ x } Wnet\ (kj/kg)}$$

$$= \frac{3600}{Wnet\ (\frac{KJ}{Kg})}$$

- It is a measure of the capacity of a steam power plant.

- Steam rate $= \frac{3600}{W_{net}} (\frac{kg}{kw - hr})$

  Where Wnet is in KJ/Kg

$$= \frac{3600}{Wnet} (\frac{kg}{kw - hr}) \text{ or } (\frac{KJ}{Kg})$$

- Higher steam rate implies, more steam consumption for particular amount of power. Hence **lower steam rate is desirable for power plant.**

Problem - 1

A steam power plant working on ideal Rankine cycle has the boiler outlet pressure of 1000 kPa and condenser pressure of 100 kPa. Find

(i) pump work (W_p) in kJ/kg;

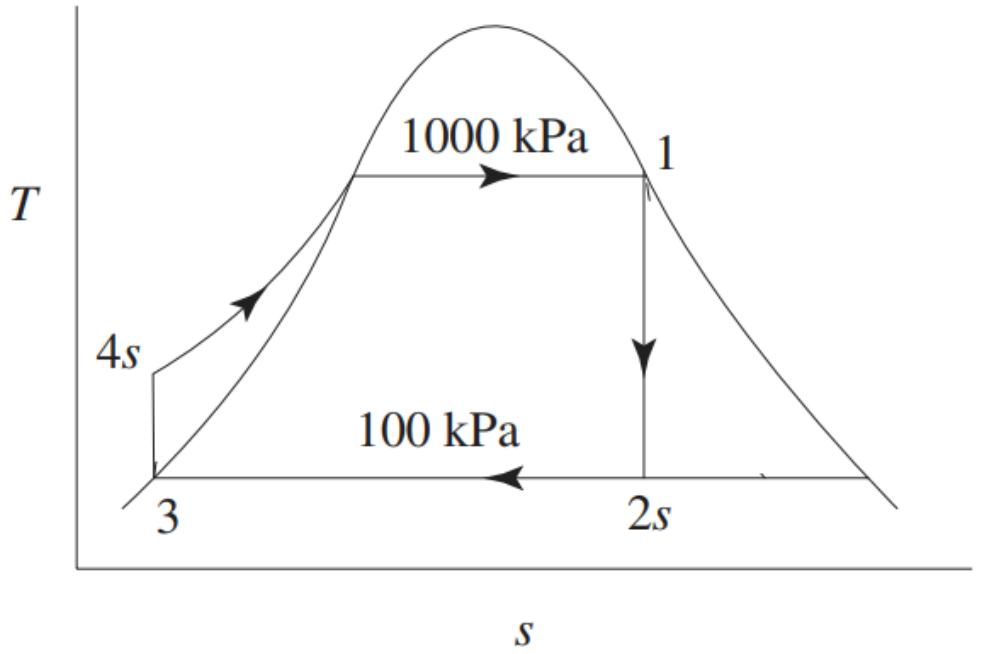
(ii) turbine output (W_T) in kJ/kg;

(iii) heat input Q in kJ/kg;

(iv) cycle efficiency;

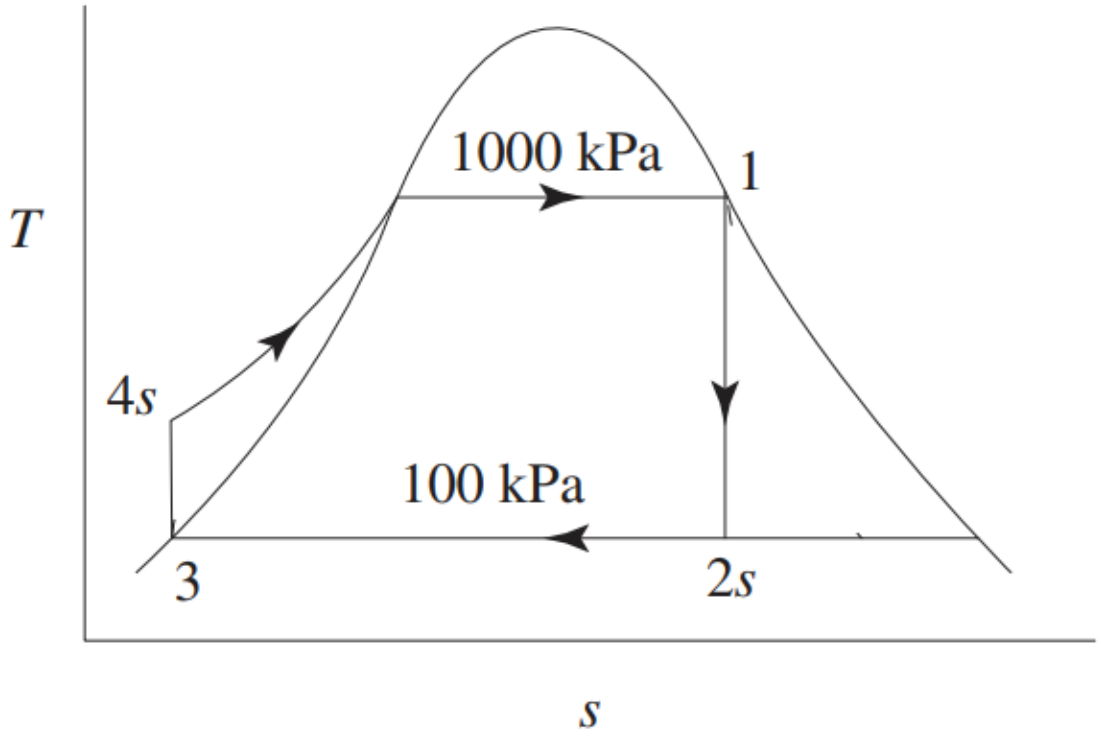
(v) steam rate in kg/kW h; and

(vi) moisture at the end of the turbine process.



Solution - 1

Referring to Fig. it is seen that the steam is expanding from 1000 kPa (1) to 100 kPa (2), i.e. from 1 MPa to 0.1 MPa. Ideal cycle is represented by 1-2s-3-4s-1. Now let us take down the various properties from the steam table at the two pressures.



$$p_1 = 1 \text{ MPa}$$

$$h_1 = h_g = 2778.1 \text{ kJ/kg}$$

$$s_1 = s_g = 6.5864 \text{ kJ/kg K}$$

$$s_1 = s_{2s} = 6.5864 \text{ kJ/kg K}$$

$$p_2 = 0.1 \text{ MPa}$$

$$h_3 = h_f = 417.16 \text{ kJ/kg}$$

$$s_3 = s_f = 1.3025 \text{ kJ/kg K}$$

$$h_{fg} = 2258.0 \text{ kJ/kg}$$

$$s_{fg_{2s}} = 6.0588 \text{ kJ/kg K}$$

$$s_{2s} = s_3 + x_{2s} \times s_{fg_{2s}}$$

$$6.5864 = 1.3025 + x_{2s} \times 6.0588$$

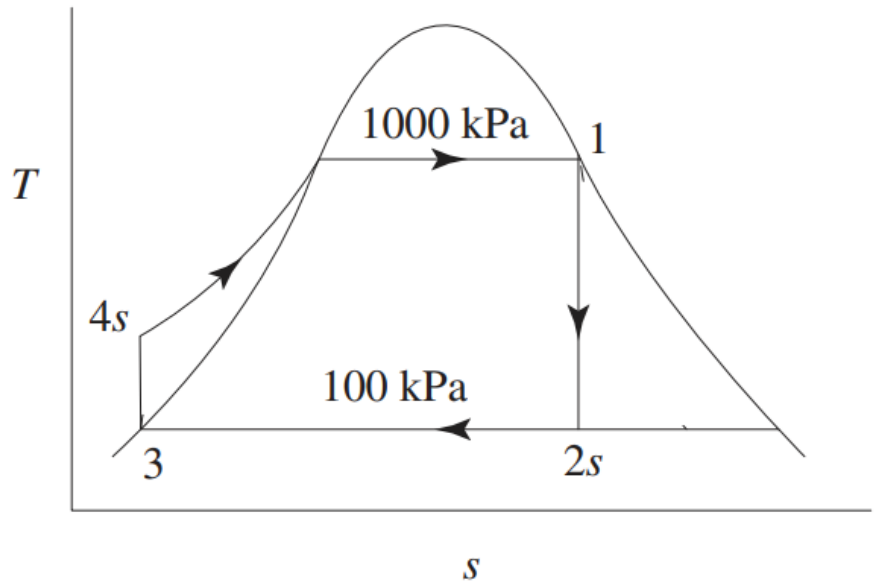
$$x_{2s} = \frac{(6.5864 - 1.3025)}{6.0588}$$

$$\boxed{x_{2s} = 0.8721}$$

$$h_{2s} = h_f + x_{2s} \times h_{fg}$$

$$h_{2s} = 417.16 + 0.8721 \times 2258$$

$$\boxed{h_{2s} = 2386.36 \text{ kJ/kg}}$$



Work input to the pump,

$$W_p = h_{4s} - h_3$$

At state 3, the working fluid is a saturated liquid.

From the steam table, the specific volume

$$v_f = 0.001 \text{ m}^3/\text{kg}$$

As liquid is incompressible, its specific volume will be constant.

$$v_3 = (v_f)_{1 \text{ bar}}$$

$$v_3 = 0.001 \text{ m}^3/\text{kg}$$

$$W_p = \int_3^{4s} v dp$$

$$W_p = v_3(p_{4s} - p_3)$$

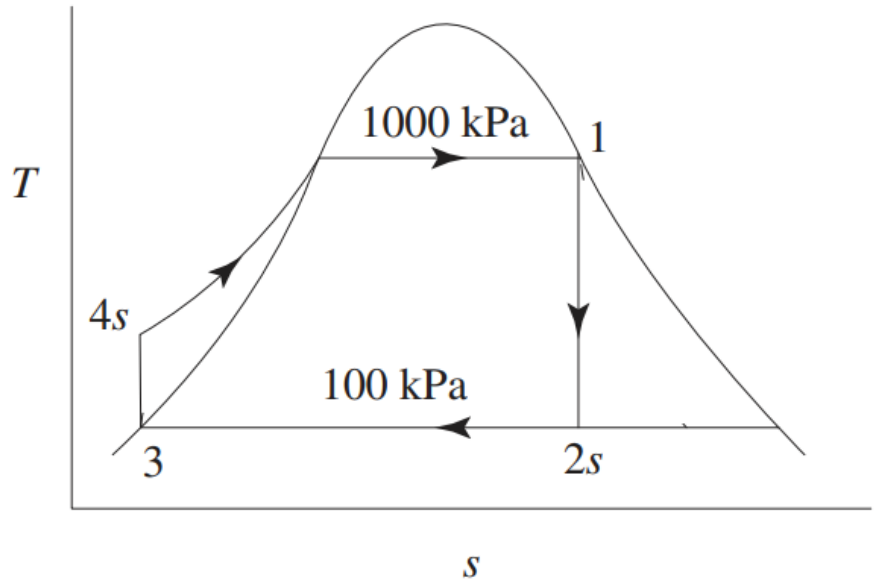
$$W_p = 0.001 \times (1000 - 100)$$

$$W_p = 0.9 \text{ kJ/kg}$$

$$\text{Turbine work, } W_T = h_1 - h_{2s}$$

$$W_T = 2778.1 - 2386.36$$

$$\boxed{W_T = 391.74 \text{ kJ/kg}}$$



$$W_{net} = W_T - W_p$$

$$W_{net} = 391.74 - 0.9$$

$$\boxed{W_{net} = 390.84 \text{ kJ/kg}}$$

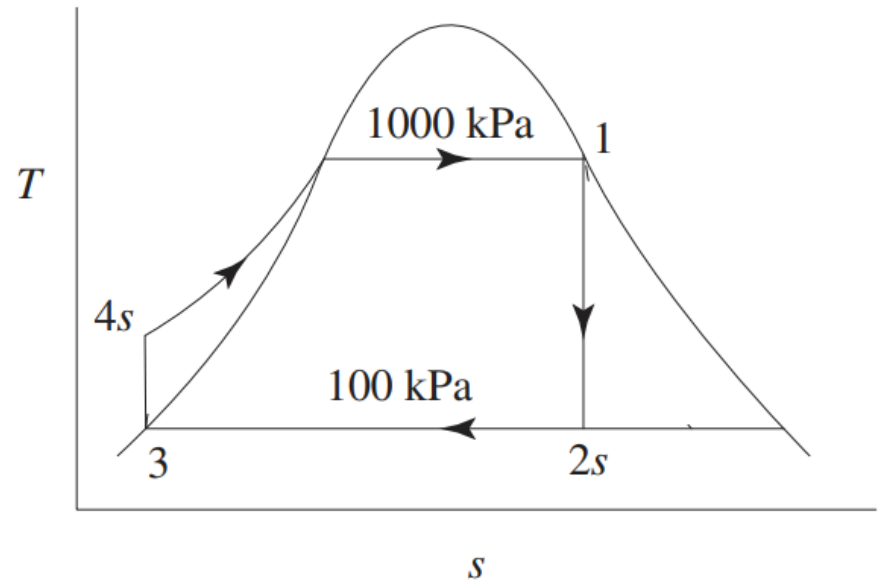
$$W_p = 0.9 \text{ kJ/kg}$$

$$W_p = h_{4s} - h_3$$

$$h_{4s} = h_3 + 0.9$$

$$h_{4s} = 417.16 + 0.9$$

$$\boxed{h_{4s} = 418.1 \text{ kJ/kg}}$$



Heat input, $Q_1 = h_1 - h_{4s}$

$$Q_1 = 2778.1 - 418.1$$

$$\boxed{Q_1 = 2360 \text{ kJ/kg}}$$

$$\eta_{cycle} = \frac{W_{net}}{Q_1}$$

$$= \frac{390.84}{2360.0}$$

$$= 0.1656$$

$$= \boxed{16.56\%}$$

$$\text{Steam rate} = \frac{3600}{W_{net}}$$

$$= \frac{3600}{390.84}$$

$$= \boxed{9.21 \text{ kg/kW h}}$$

$$\text{Moisture constant} = (1 - x_{2s})$$

$$= 1 - 0.8721$$

$$= 0.1279$$

$$= \boxed{12.79\%}$$

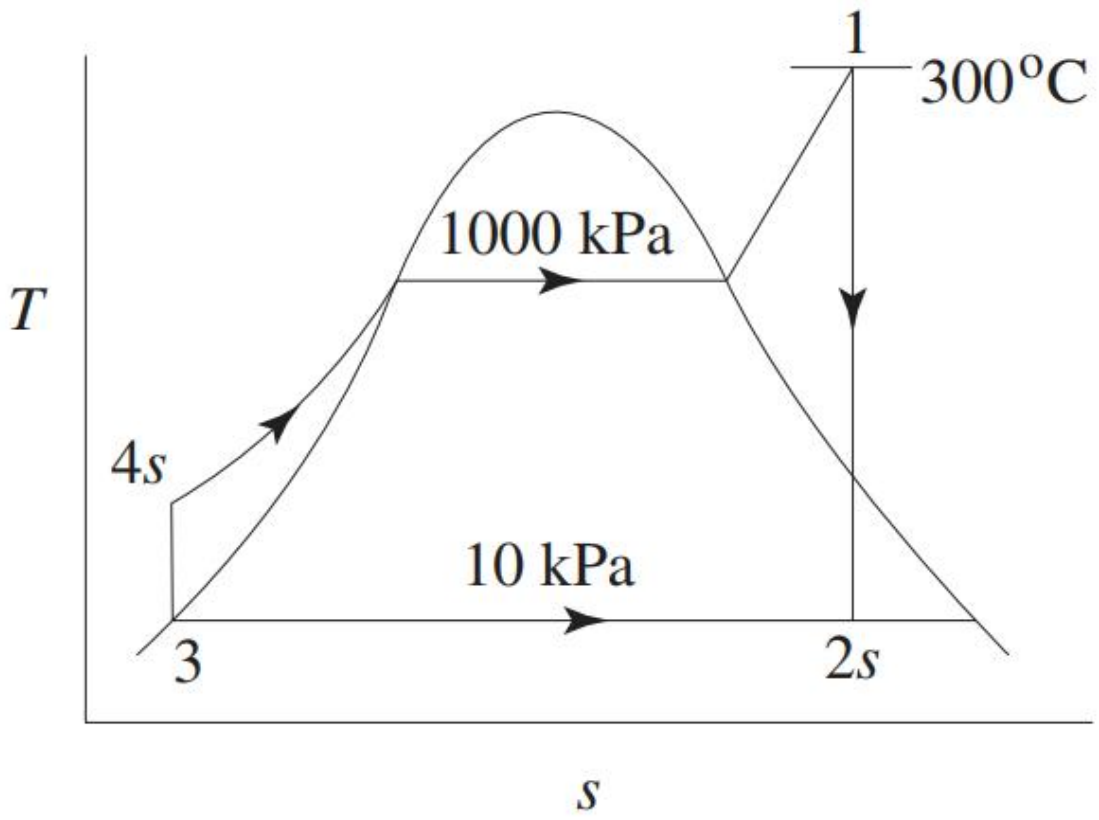
Note: Steam rate is defined as the amount of steam (in Kg/hr) required to produce unit power (in kW).

Problem - 2

In example 1, if the boiler outlet steam is superheated at 1000 kPa and 300 °C with the condenser pressure 10 kPa. Find the cycle efficiency, steam rate and moisture at turbine exit.

Solution - 2

From steam table for superheated at 300 °C and 1000 kPa and for the condenser pressure of 10 kPa, let us take down the values. Refer Fig.1



$p_1 = 1000$ kPa and $T = 300°C$ (superheat)

$h_1 = 3051.2$ kJ/kg

$s_1 = 7.1228$ kJ/kg K

$s_{2s} = s_1 = 7.1228$ kJ/kg K

$p_2 = 10$ kPa

$h_3 = h_f = 191.81$ kJ/kg

$s_3 = s_f = 0.6492$ kJ/kg K

$h_{fg} = 2392.8$ kJ/kg K

$s_{fg_{2s}} = 7.5010$ kJ/kg K

$$s_{2s} = s_3 + x_{2s} \times s_{fg_{2s}}$$

$$7.1228 = 0.6492 + x_{2s} \times 7.5010$$

$$x_{2s} = \frac{(7.1228 - 0.6492)}{7.5010} = \boxed{0.8630}$$

$$h_{2s} = h_3 + x_{2s} h_{fg} = 191.81 + 0.8630 \times 2392.8 = \boxed{2256.8 \text{ kJ/Kg}}$$

$$W_p h_{4s} = h_{4s} - h_3 = 1 \text{ kJ/kg (from previous example)}$$

$$h_{4s} = W_p + h_3 = 1 + 191.86 = \boxed{192.86 \text{ kJ/kg}}$$

$$W_T = h_1 - h_{2s} = 3051.2 - 2256.8 = \boxed{794.4 \text{ kJ/kg}}$$

$$W_{net} = W_T - W_p = 794.4 - 1 = \boxed{793.4 \text{ kJ/kg}}$$

$$Q_1 = h_1 - h_{4s} = 3051.2 - 192.86 = \boxed{2858.34 \text{ kJ/kg}}$$

$$\eta_{cycle} = \frac{W_T}{Q_1} = \frac{794.4}{2858.34} = 0.278 = \boxed{27.8\%}$$

$$\text{Steam rate} = \frac{3600}{W_{net}} = \frac{3600}{793.4} = \boxed{4.537 \text{ kg/kW h}}$$

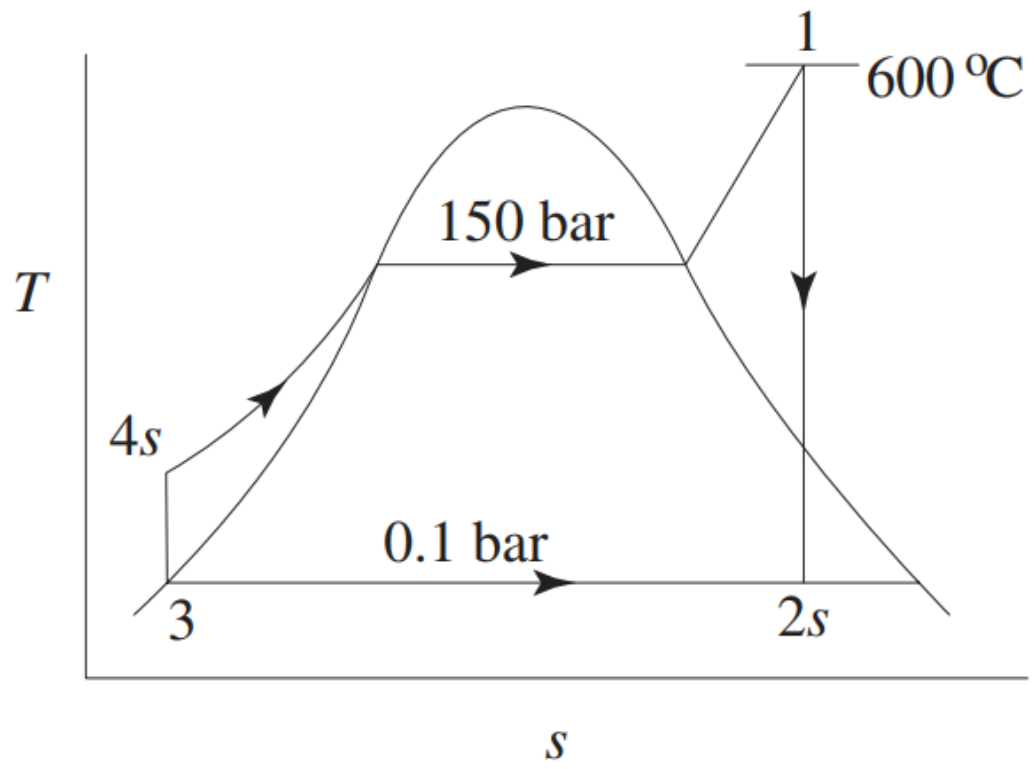
$$\text{Moisture content} = 1 - x_{2s} = 1 - 0.863 = 0.137 = \boxed{13.7\%}$$

Problem - 3

In Example 1, if the boiler outlet pressure is 150 bar at 600 °C with a condenser pressure of 10 kPa and working on ideal Rankine cycle. Calculate the cycle efficiency, steam flow rate and moisture content.

Solution - 3

Here, the boiler pressure is 15 MPa and condenser pressure is 10 kPa. Further, the steam is superheated to 600 °C. Hence, let us write down the required property values from the steam table.



$p_1 = 15$ MPa, $T_1 = 600°C$

$h_1 = 3582.3$ kJ/kg

$s_1 = 6.6775$ kJ/kg K

$s_{2s} = s_1 = 6.6775$ kJ/kg K

$p_2 = 10$ kPa

$h_3 = h_f = 191.81$ kJ/kg

$s_3 = s_f = 0.6492$ kJ/kg K

$h_{fg_{2s}} = 2392.8$ kJ/kg

$s_{fg_{2s}} = 7.5010$ kJ/kg K

$$s_{2s} = s_1 = s_3 + s_{fg_{2s}} \times x_{2s}$$

$$6.6775 = 0.6492 + 7.5010 \times x_{2s}$$

$$x_{2s} = \frac{(6.6775 - 0.6492)}{7.5010} = \boxed{0.804}$$

$$h_{2s} = h_3 + x_{2s} \times h_{fg_{2s}} = 191.81 + 0.804 \times 2392.8 = \boxed{2115.62 \text{ kJ/kg}}$$

$$h_{4s} = W_p + h_3$$

$$W_p = \int_3^{4s} v_f dp = v_f \times [p_{4s} - p_3] = 0.001 \times (15000 - 10) \approx \boxed{15 \text{ kJ/kg}}$$

$$h_{4s} = 15 + 191.81 = \boxed{206.81 \text{ kJ/kg}}$$

$$W_T = h_1 - h_{2s} = 3582.3 - 2115.62 = \boxed{1466.8 \text{ kJ/kg}}$$

$$Q_1 = h_1 - h_{4s} = 3582.3 - 206.81 = \boxed{3375.49 \text{ kJ/kg}}$$

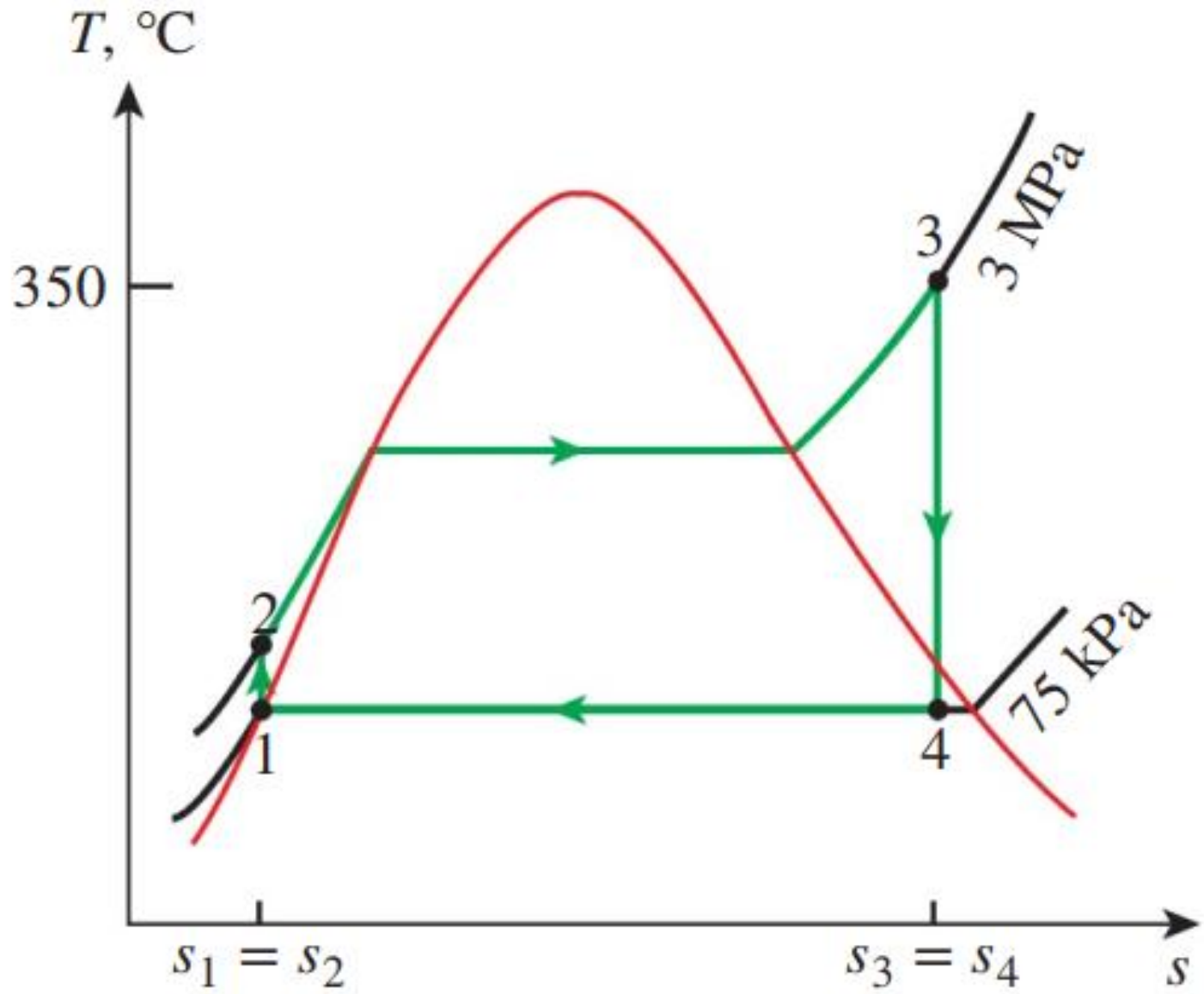
$$\eta_{cycle} = \frac{W_T}{Q_1} = \frac{1466.68}{3375.49} = 0.4345 = \boxed{43.45\%}$$

$$\text{Steam rate} = \frac{3600}{1466.68} = \boxed{2.4545 \text{ kg/kW h}}$$

$$\text{Moisture} = 1 - x_{2s} = 1 - 0.804 = 0.196 = \boxed{19.6\%}$$

Problem - 4

Consider a steam power plant operating on the simple ideal Rankine cycle. Steam enters the turbine at 3 MPa and 350° C and is condensed in the condenser at a pressure of 75 kPa. Determine the thermal efficiency of this cycle.


T, °C
350
3
3 MPa
2
1
4
75 kPa
$s_1 = s_2$
$s_3 = s_4$
s

State 1: $P_1 = 75$ kPa, Sat. liquid $\Rightarrow$ $h_1 = h_{f\,@\,75\text{ kPa}} = 384.44$ kJ/kg, $\upsilon_1 = \upsilon_{f\,@\,75\text{ kPa}} = 0.001037$ m^3/kg

State 2: $P_2 = 3$ MPa

$$s_2 = s_1$$

$$w_{\text{pump,in}} = \upsilon_1(P_2 - P_1) = (0.001037\ \text{m}^3/\text{kg})[(3000 - 75)\ \text{kPa}]\left(\frac{1\ \text{kJ}}{1\ \text{kPa}\cdot\text{m}^3}\right)$$

$$= 3.03\ \text{kJ/kg}$$

$$h_2 = h_1 + w_{\text{pump,in}} = (384.44 + 3.03)\ \text{kJ/kg} = 387.47\ \text{kJ/kg}$$

State 3: $P_3 = 3$ MPa, $T_3 = 350°\text{C}$ $\Rightarrow$ $h_3 = 3116.1$ kJ/kg, $s_3 = 6.7450$ kJ/kg·K

State 4: $P_4 = 75$ kPa (sat. mixture)

$$s_4 = s_3$$

$$x_4 = \frac{s_4 - s_f}{s_{fg}} = \frac{6.7450 - 1.2132}{6.2426} = 0.8861$$

$$h_4 = h_f + x_4 h_{fg} = 384.44 + 0.8861(2278.0) = 2403.0\ \text{kJ/kg}$$

Thus,

$$q_{in} = h_3 - h_2 = (3116.1 - 387.47)\ \text{kJ/kg} = 2728.6\ \text{kJ/kg}$$
$$q_{out} = h_4 - h_1 = (2403.0 - 384.44)\ \text{kJ/kg} = 2018.6\ \text{kJ/kg}$$

and

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{2018.6\ \text{kJ/kg}}{2728.6\ \text{kJ/kg}} = 0.260 \text{ or } \mathbf{26.0\%}$$

The thermal efficiency could also be determined from

$$w_{turb,out} = h_3 - h_4 = (3116.1 - 2403.0)\ \text{kJ/kg} = 713.1\ \text{kJ/kg}$$
$$w_{net} = w_{turb,out} - w_{pump,in} = (713.1 - 3.03)\ \text{kJ/kg} = 710.1\ \text{kJ/kg}$$

or

$$w_{net} = q_{in} - q_{out} = (2728.6 - 2018.6)\ \text{kJ/kg} = 710.0\ \text{kJ/kg}$$

and

$$\eta_{th} = \frac{w_{net}}{q_{in}} = \frac{710.0\ \text{kJ/kg}}{2728.6\ \text{kJ/kg}} = 0.260 \text{ or } \mathbf{26.0\%}$$